

What theories of consciousness commit to, and what they leave unmeasured: an audit of stated predictions after the first adversarial test

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Running head. What theories of consciousness commit to

Abstract

The first preregistered adversarial test of consciousness theories compared two theories on three divergent predictions about the cortical signature of conscious content; each passed one criterion and failed one. We ask what the theories have committed to beyond that signature, and what they have left unmeasured. A purposive audit coded the stated predictions of 12 accounts against 10 measurable quantities (120 cells), registered 27 stated predictions and 6 derived by us and classified the content manipulated in 42 experiments; column classes were derived by script, not by a coder. The result is a concentration. Only 21 of 120 cells carry a stated prediction; 6 of 12 accounts state one about the cortical signature, the one column that is contested — of 54 cross-theory pairs of stated predictions there, 2 are discriminating, 10 jointly compatible and 42 merely different. Elsewhere commitment is sparse: the number and the incompatibility of competing action policies have one explicit occupant each, and so does visceral modulation of access; interoceptive content and autonomic conflict have two each, whose predictions do not conflict; cortical–autonomic coherence has none. The empirical base mirrors this: 18 of 30 theory-addressed experiments manipulated neutral visual content, 0 of 30 valenced content. The next adversarial test needs a condition on a quantity the theories have committed to and never measured, not another contrast on the cortical signature. We specify one — incompatibility of co-executable action policies at fixed count under no-report access — that tests two operationalisations we derived (P32, P23) from one account’s stated commitments; a null bounds the derivations and refutes no theory.

Keywords: consciousness; adversarial collaboration; conscious access; theory audit; stated predictions; action-policy competition; interoception; column typology

1. Introduction

For thirty years the standard complaint about consciousness science was that its theories were not tested. The complaint has been answered, and the answer was worse. The Cogitate consortium ran a preregistered adversarial test of global neuronal workspace theory (GNWT) and integrated informa-

tion theory (IIT) across fMRI, MEG and intracranial EEG in 256 participants, with pass and fail criteria agreed in advance by proponents of both theories and by a theory-neutral consortium [1], and reported that neither central prediction survived intact [2]. GNWT was challenged by the absence of prefrontal ignition at stimulus offset — a null that cannot be blamed on insensitivity, since offset responses were robust elsewhere and the same prefrontal sites showed strong onset responses to the same stimuli. IIT was challenged by the absence of sustained within-posterior synchronisation. The consortium declined a verdict, adopting a Lakatosian soft-falsification frame, and GNWT proponents have published their own reading of the same data [3].

The usual lesson drawn is about the theories: too vague, too unfalsifiable, too well defended by auxiliary assumptions. We draw a different one, and it is not the lesson we drew in the version of this review that was first submitted. There we argued that the two theories were tested on a prediction they share and could therefore not be told apart by the experiment that tested them. That was wrong as a description of what was done. The consortium preregistered three predictions on which GNWT and IIT diverge — where conscious content is decodable, whether its neural correlate is sustained for the duration of the percept or ignites at onset and offset, and whether the connectivity that carries it is fronto-posterior or confined to posterior cortex — and each theory passed at least one criterion and failed at least one [1,2]. The test discriminated. What it discriminated on is a single quantity, the cortical signature of conscious content, and the consortium's own remark that its challenges extend to any theory sharing the predictions tested [2] is a remark about how many theories have staked their commitments on that one quantity.

That is the thesis of this review. The stated commitments of theories of consciousness concentrate on a few measurable quantities, and most of what the theories could be tested on is unmeasured — not because the measurements are unavailable but because no theory has said what it expects. The claim needs a vocabulary, since “unmeasured” covers cases that license different inferences. We classify each measurable quantity — each *column* of the audit in Section 7 — by what the coded theories have stated over it, and the class is computed from the coded cells rather than assigned by a reader. A column is *contested* when at least two accounts state predictions over it and at least one pair of those predictions is *discriminating*: both speak to a shared observable under a shared condition, and some value of it is predicted by one and excluded by the other. Descriptions that merely differ — a prediction about locus beside one about timing, or a claim about one axis of judgement beside a claim about the structure of experience — are recorded in the deposited pair register as *different* or *jointly compatible* and do not make a column contested. It is *occupied but not contested* when at least two accounts state predictions and no pair is discriminating, so an experiment can corroborate or disconfirm them but not separate them. A single stated prediction can of course fail on its own; what only a contested column offers is a result that favours one occupant over another. It is *single-occupant* when exactly one account states a prediction; a positive result confirms that account and leaves the others where they were, and the informative outcome is the null. It is *thin* when no account states a prediction but at least one is compatible with the quantity mattering, and *unoccupied* when none is. Silence is not a contrary prediction: an experiment on a single-occupant or thin column adjudicates between none of the silent theories. Which of the ten quantities coded below falls into which class is the question Section 7 answers, and Section 9 draws the design consequence.

Locating such a quantity requires knowing what the theories actually predict, as opposed to what they are said to predict in review-article summaries; that is this review's contribution. We carried out a purposive audit. The stated predictions of 12 accounts — 10 trial-level theories of conscious access and 2 origin-level accounts reported apart — were coded against 10 measurable quantities (120 cells) from primary statements and the theories' own self-profiling reviews; each stated prediction was entered in a register with its source, condition, outcome measure and expected result (27 stated by the theories, 6 derived by us and labelled as such), and every cross-theory pair of stated predictions in the columns with two or more was entered in a pair register with its relation; and the content whose access was manipulated in 42 experiments from 32 publications that these theories are tested by or built on was classified separately. The audit is not a systematic review, and it reports counts of coded accounts rather than estimates of a population; its codes are those of a single coder until the scheduled blind second coding is complete, and every count is provisional to that extent. It is evidence for a claim otherwise easy to assert and hard to check: that the field's stated predictions are concentrated on the cortical signature of access, where they are contested; that the quantities the surrounding biology treats as fundamental to arbitration and allostasis are single-occupant or thin; and that the column we had taken to be contested is, once the predictions are written down beside each other, occupied by accounts answering different questions.

Three conventions govern what follows. This is an evaluative review, not a census: every section closes with a judgement about which position the evidence supports, and where a widely repeated claim is weaker than its reputation we say so. We distinguish throughout between claims about *access* — content becoming available to many consumers, and claims about *phenomenal experience*. We use one definition of “conscious access” throughout, the one in general use: content is accessed when it is reportable and available to multiple downstream cognitive systems. We do not build any further clause into it. The proposal that access measurably alters organism state, whether or not it produces fronto-parietal ignition, is a hypothesis we advance — it is what column M5 asks about — and it is kept outside the definition so that the coding of M5 cannot be definitional: each M5 cell asks whether *that theory* states a prediction over organism-wide cortical-autonomic coherence, which is a question about its predictive apparatus and not about how access is defined. Among the 12 coded accounts, 5 are coded INTERPRETED there because organism-wide integration is in their spirit but is operationalised cortically, through a single visceral channel, or not at all, and 7 NOT_LOCATED because no statement bearing on the quantity was found in their primary sources. Neither coding depends on the definition of access. What our hypothesis affects is how much a thin column *matters*, and that is an argument we make openly rather than smuggle through a definition; Section 7.4 also states why the heartbeat-evoked-response literature, which does couple a cortical measure to a visceral signal, is coded as compatible with M5 rather than as a stated prediction over it, and why M5 is a construct we propose, not part of the definition of access. Where a formulation carrying neither commitment is needed we use *global availability*, in the descriptive sense of content being available to many consumers rather than in the theoretical sense the term has inside global workspace theory [4].

The review proceeds from conceptual bookkeeping to measurement. Section 2 asks which distinctions the field has earned. Sections 3 to 5 evaluate the three literatures supplying the raw material for any account of why an organism would need conscious access at all: biological self-maintenance and basal cognition, action arbitration, and valence as organism-wide reconfiguration. Section 6 maps the

disputes between theories and what the adversarial test did and did not decide; Section 7 presents the audit and reports the class of each coded quantity as the coding returned it; Section 8 examines three unmeasured dissociations; Section 9 states the minimal measurement that follows from the classification, and says exactly which stated prediction it tests. A theory making predictions about the single-occupant and thin coordinates exists and is presented separately; we name it once, at the end, and do not argue for it here.

2. Earned and unearned distinctions

Any claim about consciousness is a claim about one of at least seven things, and most disputes here are about which one. The coding in Section 7 depends on holding them apart: unconscious processing; conscious content; access; online metarepresentation of one's own processing; control; post-hoc reconstruction; the putative bearer of phenomenal states. These are not degrees of one variable but dissociate pairwise.

The oldest distinction is also the least stable. Block's separation of phenomenal from access consciousness [5] made the inference from "globally available" to "felt" a step requiring argument, later given empirical teeth by the overflow claim that experience contains more than the system can report [6,7]. The programme that grew from it — isolating the correlates of conscious content from its prerequisites and consequences [8], and distinguishing conscious state from content [9] — is a genuine achievement, and the no-report literature of Section 6 its most valuable product.

What has changed is confidence in the two-kinds framing. An argument from inside the experimental camp holds that dividing consciousness into two *kinds* converts a graded multi-component phenomenon into two entities, one inaccessible to measurement by construction [10]. That is a conceptual paper, not a result, but it converges with an independent movement toward feature profiles: multidimensional assessment of animal sentience replacing a single scale [11], indicator lists for invertebrates [12], a framework treating consciousness as a profile of separately assessable capacities [13].

The field has earned the process/content/access distinctions and has *not* earned the phenomenal/access dichotomy in its strong form, and the two are usually run together to the detriment of both. The first set is earned because each distinction is independently measurable and has produced dissociations that survived attempts to explain them away: integration without awareness [14], frontal correlates tracking report rather than the percept [15], conscious state varying independently of content [9]. The dichotomy is not, because its central claim — a second kind of consciousness whose presence comes apart from access — is what no measurement can establish; overflow arguments turn on the interpretation of partial report, the opposing position denies the separation is coherent [16], an intermediate view treats access as partial and hierarchical rather than all-or-none [17].

The cost of conflating these levels shows up in how theories get evaluated. A theory specifying when content becomes globally available answers an access question, and its success says nothing directly about phenomenality; but with the dichotomy in the background such theories are read as accounts of phenomenal experience, criticised for failing a task they never took on. The converse error is as common: theories staking a claim on phenomenality are credited with the access evidence. Feature

profiles help by forcing each claim to name its own observable, which is what makes the coding of Section 7 possible.

3. From regulation to a minimal bearer

The literature on biological self-maintenance rigorously accounts for what makes a boundary an organism's own; what it does not give, and largely does not claim to, is any route from organisation to a felt aspect.

Autopoietic theory describes a living system as a network of processes producing its own boundary, treating this closure as sufficient for "cognition" [18]. Di Paolo's refinement is decisive: bare autopoiesis is insufficient even for agency, since a system that either maintains its organisation or disintegrates has no gradations of better and worse and so no norms; what is needed is adaptivity, regulating one's own relation to viability boundaries [19]. **Neither text posits a felt aspect.**

The corresponding corpus is basal cognition [20]. A parallel line proposes the skin-brain thesis: early nervous systems arose not as input-output machinery but to coordinate whole-body contraction [21]. That places single-body coordination at the origin of nervous systems — Section 4's starting point.

The split is over whether regulation entails experience. A moderate position keeps the functional vocabulary: cellular and bioelectric networks as problem-solvers in morphospace, the self's computational boundary set by the scale of coordination [22]. A radical position takes the step the moderate one declines, deriving sentience from cellular organisation and locating the first minds in single cells [23] — a redefinition rather than a proposal, adding no mechanism.

Three ladders locate the threshold differently. Feinberg and Mallatt place it in the Cambrian, tied to isomorphic neural representations [24]. Ginsburg and Jablonka propose unlimited associative learning as a *marker*: a detectable sign not a definition [25]. The third is homeostatic: feeling derived from regulation of the internal milieu [26], affect localised subcortically [27], a free-energy formalisation [28].

The anti-corticocentric line pushes the shared claim — the threshold is not the presence or absence of a cortex — lower than most expect. Merker places the core of the conscious state in upper brainstem and mesodiencephalon, where body schema, action space and motivational ranking integrate, cortex supplying content rather than consciousness [29]. This consolidated into the New York Declaration on Animal Consciousness [30].

The transition from organisation to experience is warranted nowhere here and claimed seriously nowhere except in the radical cellular position — a strength rather than an embarrassment, since the basal cognition programme is valuable precisely because it stops short. Where the inference *is* warranted is narrower: from organisation to a *control problem* whose solution needs something the organisation does not contain. Coordinating many specialised subsystems around one body is such a problem, solved by nervous systems from their origin [21], and it is where the ladders converge. That convergence is the difficulty: because three independent programmes reach it, "integration around a single body matters" is not an available contribution. The unoccupied position is narrower — the *number and heterogeneity of conflicting domains* integrated is a measurable variable — and no ladder

contains that axis, though transitions in cognitive evolution are described as changes in control regime [31].

4. Arbitration as a function

If conscious access exists because incompatible demands from many specialised systems must be reduced to one action of one body, the mechanisms performing that reduction are where to look. Single-body arbitration has been solved unconsciously three times over, in three formalisms, none requiring access.

The first solution is anatomical. The basal ganglia were proposed as the vertebrate solution to the selection problem: many command systems compete for one body's skeletal motor apparatus, and a centralised device is needed to resolve the conflict without being one of the competitors [32]. Formalisation showed striatal, subthalamic and pallidal microanatomy implementing an off-centre on-surround scheme [33]. This arbiter operates entirely without access [34].

The second solution is computational. Bounded accumulation of evidence to a threshold reproduces reaction-time distributions and accuracy jointly, parietal activity tracking the integral of evidence for an option [35].

The third solution is architectural. The affordance competition hypothesis holds that the visual system specifies several available actions at once while fronto-parietal circuits maintain them as mutually inhibiting population representations biased by value [36]. With two permissible movement directions, monkey dorsal premotor cortex carries two directional activity peaks simultaneously, and only after a cue is one suppressed and the other amplified to threshold [37].

Supramodular interaction theory (SIT) holds that phenomenal states exist to permit interaction among supramodular response systems — agentic, encapsulated structures defined by concerns rather than input modality — which compete for the same skeletomotor output and which, without phenomenal states, could not influence action *jointly* [38]. The passive frame formulation narrows the claim to the somatic nervous system, the mechanism passive and low-level [39].

SIT's empirical base is of three unequal kinds, stronger than its critics allow. The first is taxonomic: introspectively penetrable conflicts (pain for gain, breath holding, delay of gratification) are separated from impenetrable ones (pupillary reflex, ventriloquism and McGurk effects), the dividing line running through a skeletomotor plan. The second is experimental: participants trained to introspect rated the urge to err while sustaining compatible versus incompatible skeletomotor intentions, incompatible intentions producing stronger subjective effects while a smooth-muscle control produced none [40]. That control carries the load: without it the theory would predict a subjective effect from any antagonism. The measure in both conditions is a rating of experienced conflict. It shows that incompatible skeletomotor intentions are accompanied by a reportable state; it does not show that the arbitration required that state, nor that the state is conscious access to anything beyond the conflict itself. Arbitration is demonstrated by the outcome of the competition, access by a separate measure, and the experiment supplies the first and reports the second. Section 9 accordingly treats the step from experienced conflict to access as a derivation of ours, labelled as such, rather than as a claim of the theory. The third support is neuroimaging and markedly weaker: subjective

conflict was associated with left postcentral, that is somatosensory rather than executive, cortex [41]. The sharpest question is Clark's: did voluntary skeletomotor action arise for control of the external environment or the internal one, with voluntary regulation of internal-state conflicts as a source of visceral awareness [42].

Our judgement has two parts. First, the three unconscious solutions establish decisively that policy competition is *insufficient* for access: competition is present in all three and access is absent. They establish nothing about its necessity. Necessity would require showing that access does not arise without policy competition, and nothing reviewed here bears on that; and the existence of arbitration, however well demonstrated, establishes nothing about consciousness in either direction, since the three unconscious arbiters show that arbitration proceeds without it and no experiment reviewed here shows that access changes when arbitration does. §9 accordingly tests the one stated prediction over the arbitration situation — the supramodular threshold — together with its extension to access, which no theory has stated. Any theory claiming that access exists *for* arbitration therefore owes an account of what distinguishes the arbitration requiring access from the arbitration that does not; pointing at the existence of conflict is not that account.

Second, SIT does supply such an account, and it is *taxonomic* rather than computational: the conflict either sits at the action-selection stage and involves skeletal musculature, or it does not. This is the strongest available answer, and it pays a price: the classifying feature says where the conflict is located but not what makes it unresolvable without access, so the explanation cannot transfer without re-running the taxonomy. A related cost attaches to the passive frame formulation: the conscious field presents contents but does not select, and operates whether or not conflict is present [43]. But a frame passive with respect to selection has no variable that can differ between arbitration problems, as a discriminating prediction requires. SIT buys defensibility with a commitment hard to extend, and the cell where an extension would be tested — non-skeletal conflict — is where it states a null: conflict within a smooth-muscle effector, with no skeletomotor plan, produces no experienced conflict [40]. The null is an exclusion of autonomic conflict from the function of the conscious field, not a denial that bodily states are felt — the same programme names tooth pain and urges among the most basic conscious contents, and its authors declined an invitation to extend the theory to the autonomic nervous system rather than deny that autonomic events are experienced [39]. Section 7.4 records what that distinction does to the column it sits in.

5. Valence and organism-wide reconfiguration

An arbitration account needs a common scale. The argument that incommensurable motives must become a single quantity for behaviour to be selectable, with pleasure as that currency, was made decades ago [44], and neuroeconomics gave it a neural reading as a common currency of subjective value [45]. The problem is posed *independently of theories of consciousness*, so a theory of access saying something about the scale makes a claim none of its competitors makes.

The interoceptive literature has meanwhile moved the body from monitored object to control loop, from afferent representation of homeostatic state in insula [46]. The decisive step was redescribing it as *inference* rather than readout [47], regulation proceeding by autonomic, endocrine and behavioural action bringing the body to the predicted state [48].

Organism-wide scope comes from allostasis rather than interoception as such: set points change in advance on anticipated need [49]. Functional connectivity identified a large-scale network named allostatic-interoceptive [50]. Two caveats are load-bearing: that network came from resting-state and meta-analytic methods, not causal manipulation, so it is architectural plausibility rather than demonstrated mechanism; and a recent treatment separates interoception as *modelling* the body from allostasis as *controlling* it [51].

The thesis that an emotion brings many subsystems into a compatible regime is well supported and already owned three times over. Component process theory defines emotion as synchronisation of organismic subsystems [52]. The evolutionary tradition describes emotions as operating modes tuning many parameters for a situation type [53]. Emotion psychology adds control precedence: emotion seizes control and reconfigures action readiness rather than adding content [54]; the constructionist line says the same in allostatic vocabulary [55].

The empirical base is weaker than the theoretical statements suggest. Neuroimaging meta-analysis found no category-specific brain signatures for discrete emotions [56]; meta-analysis of autonomic patterns found the same for the periphery [57]. The two nulls do not show that global reconfiguration occurs; they show that if it occurs it takes no reproducible configuration per category. The evidence that it occurs is the evidence of the previous paragraph — synchronisation of subsystems, control precedence, operating modes — and that evidence is theoretical statement and behavioural observation rather than a measured organism-wide state. What the nulls do settle is the unit of analysis: not a category with a signature but a population of states sharing a functional task [58]. Formal models deriving valence from regulatory dynamics rather than positing it — the rate of change of prediction error [59] — are proposals rather than results, but the closest formal analogue available.

Two refinements are non-optional. First, valence is not one parameter: hedonic “liking” and motivational “wanting” dissociate neurochemically and anatomically [60], so a one-dimensional good/bad axis coarsens what is measured. Second, controlling and experienced valence must be separated. The two-system framework holds that subcortical survival circuits produce defensive responses and bodily mobilisation while experienced fear is a separate cortical product [61]; in the stronger version, experienced emotion requires higher-order representation of the state as one’s own [62], resting on the priority of metarepresentation in conscious access [63].

The LeDoux and Brown critique is correct as a constraint and overextended as a conclusion. It is correct that an integrated valent configuration can exist without being experienced, so no account may treat organism-wide valent integration as sufficient for a putative bearer of phenomenal states. It is overextended in inferring that experienced valence requires the higher-order architecture proposed: the step from “integration is not sufficient” to “metarepresentation is what is missing” is not supported by the dissociation data — the affect-first side supplies a rival subcortical mechanism [27,28]. The pain literature is the best model for keeping three levels apart. Nociception is phylogenetically broad and uncontroversial [64]; operational criteria for sentience rest not on it but on features requiring integration — motivational priority shifts, trade-offs, self-administered analgesia [65].

The judgement is that the middle level is real, measurable and under-theorised. Global defensive or appetitive reconfiguration is measurable through motivational priority and trade-off behaviour rather than report, **though it has not yet been measured as a joint cortical–autonomic state on a trial, which is the sense in which Section 7 calls the corresponding column thin**, and is where the arbitration and valence literatures meet. What no one has done is make the *structure* of that configuration a function of the *structure* of the conflict resolved: allostatic and constructionist accounts derive affect from regulation and prediction error, neuroeconomics assumes the scale, SIT has no affective variable. The proposition that the number and heterogeneity of conflicting domains sets the required dimensionality of valent integration appears unclaimed — and unlike most such propositions, it is measurable.

6. Theories and their empirical tests

The theories are easier to describe as a list of disagreements than of schools. Almost everyone agrees that conscious content is connected to broad availability; the disagreements are about what makes it conscious — network topology, causal structure, second-order representation, a predictive model of the body. We organise the map by dispute, where the empirical blows of the last five years landed, most striking several theories at once.

6.1 Where content lives

The localisation dispute — posterior versus prefrontal — was the most productive, because both sides stated mutually exclusive predictions. The workspace line ties access to late global ignition in a network privileging prefrontal cortex [66,67], its strongest evidence nonlinearity: a second attentional-blink target’s fate is bimodal rather than graded [68], and the masking threshold coincides with a sharp inflection in late components [69]. Notably, GNWT’s rationale for broadcast is flexibility rather than arbitration — an answer to the frame problem [70], a serial router for composite mental programmes [71]. IIT proceeds from phenomenological axioms to maximally irreducible integrated information, placing the hot zone in posterior temporo-parieto-occipital cortex [72,73], experience identified with the distinctions and relations unfolded from a substrate [74]. Recurrent processing theory holds instead that local recurrent activity in sensory cortex makes content conscious, entailing phenomenal content without access [75,76]. The dispute was long reproduced positionally [77], with the well-founded suspicion that outcomes depend on method and on who runs the experiment [78].

6.2 Who predicts phenomenality

Few theories predict phenomenality rather than access, and the price is metaphysical load or denial of the explanandum. **The claim concerns predictive content, not the number of frameworks that address phenomenality. Kuhn’s [79] landscape arrays theories of consciousness by ontological commitment — materialist positions in several subcategories, non-reductive physicalism, quantum theories, integrated information, panpsychisms, monisms, dualisms and idealisms — each self-described by its adherents, with critique kept to clarification and no attempt to adjudicate among them; its organising axis is what a theory says consciousness is, not which observable it stakes. The inclusion rule of Section 7 cuts across that landscape: a framework enters Table 1 only if it states a prediction over a measurable quantity, so whole regions of Kuhn’s map contribute no row — not**

because they are mistaken but because they answer a different question. Poznanski [80], for example, proposes a constitutive account on which conscious actuality is the intrinsic realisation of a transscale ontophysical continuity; the framework addresses phenomenality at the ontological level and, as presented, states no measurable prediction, and so falls outside the inclusion rule of Section 7. Higher-order theories predict structurally: being conscious is constituted by a second-order representation, entailing dissociations of metacognitive sensitivity with first-order processing intact [63,81]; the most operationalised variant nests a one-dimensional awareness state above a high-dimensional perceptual space [82], with consequences for confusing perception and imagination [83]. Attention schema theory is more radical: a system attributes awareness to itself because it models its own attention [84,85], with an attempt to reconcile AST, GNW, HOT and illusionism [86]. Predictive processing explains not the presence of experience but its structure — perception as controlled hypothesis [87], interoceptive inference as the basis of emotion and bodily selfhood [47,88] — claiming only a basis for correlate-hunting [89]. The poles bound the space: the hard problem as non-derivability of experience from function [90], the meta-problem with illusionism as its answer [91]; the working stake is more modest, making the hard problem operationally easier through accumulated structural correspondences [92].

The methodological threat is logical rather than empirical. The unfolding argument shows that if consciousness is defined by causal structure, a functionally equivalent unfolded network without recurrence behaves identically, including in all reports about experience, while receiving a different value of the measure — so the theory is either false or indistinguishable from its opposite by any behavioural experiment [93]. Responses divided between rejecting its behaviourism [94] and noting a hidden premise about brain states [95]. The problem is deeper than any one theory: falsification requires independent access to phenomenality, and without that bridging assumption every theory is protected from refutation [96] — hence criteria for empirical theories [97], a Lakatosian accounting scheme [98], separating a testable weak core from a loaded strong one [99]. The 2023–2025 controversy over whether IIT is pseudoscientific continues the same dispute, and we report it without taking a side [100,101].

6.3 Access without report

Dependence on report biases toward frontal theories, and the key result is direct: in binocular rivalry, frontal activity tracks introspection and the act of reporting rather than the percept [15]. The programmatic statement [102] and the isolation of perceptual from report-related correlates [103] extend to a report-independent bifurcation signature [104], large-scale networks in a no-report visual paradigm [105] and audition [106]. Calibration is needed both ways: report reliably adds frontal correlates, but null frontal results do not themselves bury prefrontal theories [107], and no-report designs have their own defect — without report one does not know what the participant experienced [108].

6.4 What Cogitate decided, and what it left open

Exactness matters, because the loose version circulates widely. The consortium preregistered five points of divergence between GNWT and IIT and tested three of them as critical: the locus of decodable conscious content (prefrontal and parietal cortex in tandem with high-level sensory cortex, against the posterior hot zone), the temporal profile of its neural correlate (a transient ignition at

stimulus onset and again at offset, against activity sustained for the duration of the percept), and the connectivity that carries it (fronto-posterior synchronisation, against synchronisation within posterior cortex) [1,2]. Each theory passed at least one criterion and failed at least one. For IIT the most direct challenge was the absence of sustained within-posterior synchronisation; its content-maintenance prediction passed the pre-set criteria — sustained responses reflecting stimulus duration appeared in occipital and lateral temporal cortex — but sustained representation of orientation did not, although orientation decoded reliably in all three modalities. For GNWT the principal failure was the absence of ignition at stimulus offset, with a secondary challenge on content: category decoded in prefrontal cortex independently of task, identity not at all, orientation only in MEG — raising the question whether prefrontal cortex broadcasts all conscious content or only an abstract part [2].

The adversarial test therefore did what an adversarial test is for. Predictions were fixed in advance, criteria were agreed by opponents, results were reported against them rather than reinterpreted afterwards, and the results separated the two theories on the value of a quantity both had staked a claim on: the same measurement favoured IIT on duration and disfavoured it on posterior synchrony, and favoured GNWT on prefrontal category decoding while disavouring it on offset ignition. That is discrimination, and the earlier version of this review was wrong to describe the outcome as a test that could challenge many theories and separate none. What the test did not do is decide anything outside the quantity it varied. The consortium's remark that the challenges extend to theories sharing the predictions tested [2] is exact — recurrent processing theory shares the posterior locus, and higher-order theories that treat prefrontal cortex as a supplier of content share the prefrontal one — but it reaches them only on that quantity. Whether the same theories have committed themselves to anything else, and whether any two have committed themselves to opposed values of anything else, is a question about their stated predictions, and it is answerable by writing them down side by side. Section 7 does that.

7. The measurement gap

7.1 What is coded and why

Theories of consciousness are compared, almost exclusively, on where and when conscious contents are encoded. Section 6 traced the consequence: the first adversarial test separated its two targets on the locus, timing and connectivity of one quantity, and its challenges extend to other theories sharing the predictions tested [2]. To establish how far stated commitments extend beyond that quantity, rather than assert it, we carried out a purposive audit: the stated predictions of 12 accounts were coded against 10 measurable quantities from primary statements and self-profiling reviews (Table 1; 120 cells), each stated prediction was entered in a register that records its source, condition, outcome measure and expected result (Table S4; 27 stated and 6 derived by us), with a pair register recording the relation between every cross-theory pair of stated predictions in a column, and the content whose access was manipulated in 42 experiments from 32 publications these theories are tested by or built on was classified (Table 2). The word *purposive* is doing work. The audit is not a systematic review; there was no registered search, no PRISMA flow and, at the time of writing, one completed coder

(Section 7.2). Its tallies are counts of coded accounts and are reported as such, never as fractions of a population of theories.

The theory set was assembled under one inclusion rule and one coverage aim. The rule: an account qualifies if it has at least one preregistered or explicitly stated empirical prediction about conscious access in the current literature, which excludes purely metaphysical and purely descriptive frameworks. The aim: to span the theory families against which the field's empirical corpus is organised — the four leading theories of the ConTraSt database [78], GNWT, IIT, recurrent processing theory and higher-order theory, together with the attention schema theory — and, in addition, every family we could identify that states a prediction over one of the quantities M1–M5 that the corpus does not track: the supramodular family (supramodular interaction theory and its passive-frame successor), the interoceptive-inference family (predictive processing and the beast-machine account, and the neural subjective frame programme), and the two origin-level accounts, unlimited associative learning and the Feinberg–Mallatt scheme. The rule does not pick out these rows and no others. Dendritic integration theory, which predicts that conscious perception depends on coupling between apical and basal compartments of layer-5 pyramidal neurons [109], a prediction tested under anaesthesia [110]; the temporo-spatial theory of consciousness, which states neurophenomenal hypotheses about pre-stimulus temporo-spatial alignment [111,112]; and active inference treated as an account distinct from predictive processing, which has been given a formal workspace model with behavioural and neural predictions [113], all meet the rule and were not coded. We list them so that the omission is visible and so that a second coder can add them; we do not predict here which columns they would occupy, since that is a matter for coding rather than assertion.

Two rows of the earlier version of this table are reported apart. Unlimited associative learning [114,115] and the Feinberg–Mallatt scheme [24] answer when, and in which lineages, consciousness appeared; they are not accounts of what makes a particular content accessible on a particular trial. Coding them silent on the cortical signature of access records a difference of subject matter, not a hole in a predictive apparatus, and including them in the same tallies as the trial-level accounts inflates exactly the unevenness this review argues for. They therefore form their own block in Table 1, with their own tally. Two pairs of rows that were previously merged are now separated. Supramodular interaction theory and passive frame theory share an author and a programme, but Section 4 argues that the passive-frame formulation gives up the variable a discriminating prediction would need; if the difference matters there it must be visible here, and the two are coded from their own statements. Higher-order thought theory and higher-order state space theory are likewise two rows: the state-space formulation is offered first as a model of *reports* of awareness, treats higher-order inference as necessary but possibly not sufficient, factorises a one-dimensional presence–absence state from perceptual content and predicts prefrontal involvement in decisions about stimulus absence [82], none of which the classical formulation states [63,81]. Each is coded on its own sources, and every cell of the two rows is provisional until the second coding confirms the split.

The quantities were not chosen for neutrality, and claiming otherwise would be indefensible: three of them (M1, M2, M5) are coordinates of the theory named at the end of this paper. The selection principle was different and is stated here so it can be argued with. Each quantity is a variable that the literatures reviewed in §§3–5 already treat as central to biological regulation, action arbitration and

allostasis — not a variable invented for this survey. The test is then whether any theory of *consciousness* predicts over variables that the surrounding biology treats as fundamental. That a coordinate is also an AFT predictor is a consequence of AFT having been built from those same literatures, not evidence that the column set was rigged; but a column set chosen by the first author cannot certify its own independence, and the emptiness of a column relative to it is not evidence of anything until the column set has been checked against one the first author did not choose. Section 7.5 reports the first such check. The ten quantities: three concern the arbitration situation (M1 the number of simultaneously competing action policies, M2 their degree of incompatibility, M5 organism-wide coherence across cortical and autonomic channels); four the content (M3 valence, and the three questions into which the earlier single column “autonomic involvement” has been split because the predictions coded under it answered three different questions — M4a whether interoceptive signals can themselves become the accessed content, M4b whether the state or phase of an autonomic signal modulates access to *exteroceptive* content, M4c whether a conflict resolved wholly by autonomic effectors, with no skeletomotor plan, involves conscious experience); and three the format and signature of access (M6 dimensionality of phenomenal space, M7 precision of metarepresentation, M8 cortical signature). The split of M4 is the largest change since submission, and Section 7.4 reports what it did to the column the earlier version called contested.

7.2 Coding procedure

Each cell was coded with one of five codes defined in the deposited codebook (v2, frozen at version and hash). EXPLICIT: a sentence or preregistered prediction can be quoted that names the quantity and says what should happen to access when it varies. INTERPRETED: the prediction follows from the stated mechanism in one nameable step but is not stated for this quantity — the code the earlier version called IMPLICIT. NOT_LOCATED: no statement bearing on the quantity was found in the examined sources; it is not a prediction of no effect and is never counted as one. NOT_APPLICABLE: the theory places the quantity outside its scope, and the scope statement is quoted. UNRESOLVED: sources of the same account conflict; the cell goes to adjudication. Every EXPLICIT and INTERPRETED cell carries a polarity — positive, null or negative — and a null is a prediction: a stated claim that the quantity makes no difference, or an exclusion of it from the theory’s function, is coded EXPLICIT with null polarity and reported apart from the positive cells throughout, because a reader who sees “two accounts predict M4c” will take the two to agree that the quantity matters, and in this table both predict that it does not. Each coded cell is anchored to a source, a locus, an evidence status (full text or abstract only) and a coder confidence from 1 to 3. The INTERPRETED/NOT_LOCATED boundary carries the analytic weight and we applied it conservatively, so the count of NOT_LOCATED cells is a lower bound on the field’s silence.

The earlier table’s 96 cells carry codes under the previous four-code scheme (48 NO, 26 IMPLICIT, 20 YES, 2 YES (negative)); these were mapped — YES to EXPLICIT positive, YES (negative) to EXPLICIT null, IMPLICIT to INTERPRETED, NO to NOT_LOCATED or, where the source states a scope exclusion, NOT_APPLICABLE — and not recoded, with two exceptions on the split higher-order rows (HOT × M6 and HOSS × M3, where the legacy code rested on the other account’s source and the cell is entered NOT_LOCATED and flagged); the legacy code stands beside the current one in the deposited table — 84 cells in the cell-level file and the 12 cells of the superseded autonomic column in the long-form file. The 12 legacy cells of the single “autonomic involvement” column are

superseded by the 36 cells of M4a, M4b and M4c, all coded from the sources afresh, and enter no count; the two rows added in revision were coded from their own sources. The result: 21 EXPLICIT cells (19 positive, 2 stated null), 29 INTERPRETED (22 positive, 7 stated null), 68 NOT_LOCATED, 2 NOT_APPLICABLE and 0 UNRESOLVED. Fifty cells are provisional (every cell of the M4a–M4c columns and of the separated HOT and HOSS rows) and 16 are flagged for adjudication with a written reason, the first of them predictive processing $\times$ M4c, where the mechanism is stated but the condition is not named. Five cells were re-coded by the first coder on 24 September 2026 after a second pass and carry the previous code beside the current one: two (HOSS $\times$ M4a, HOSS $\times$ M4c) had inherited the merged higher-order row’s code although the cell’s own note said the HOSS source contains no such statement; one (predictive processing $\times$ M4b) had been kept EXPLICIT against the boundary rule the note itself applied; two (Feinberg–Mallatt $\times$ M4c, predictive processing $\times$ M2) went from INTERPRETED to NOT_LOCATED on a full-text reading that found no passage from which the prediction follows in one step. The evidence base is uneven: 63 cells rest on full text and 57 on abstracts, and 27 of the 50 EXPLICIT and INTERPRETED cells — the cells that carry a code at all — rest on abstracts alone (Section 7.7); a full-text pass over the flagged cells on 24 September 2026 reached 3 of 12 sources, the rest being closed to us, and the codes that still rest on abstracts are marked provisional in the table.

The column class is not coded. It is derived by script from Table S1, the register and a pair register after coding: *contested* if at least two EXPLICIT cells and at least one cross-theory pair of stated predictions whose relation is *discriminating* or *incompatible* — a shared observable under a shared condition, with a value predicted by one and excluded by the other; *single-occupant* if exactly one EXPLICIT cell; *occupied but not contested* if at least two and no such pair; *thin* if none but at least one INTERPRETED; *unoccupied* if none. Every cross-theory pair in a column with two or more stated predictions is entered in the pair register with its relation (different, jointly compatible, discriminating, incompatible), the result that would favour each side, and a pair-level confidence from 1 to 3; the class is reported over all pairs and over pairs at confidence ≥ 2 . The script was run only after the first coding was frozen.

The first coding was carried out by the first author. A blind second coding by the second author is scheduled against the frozen codebook (version 2.2; the hash of the file above its footer is printed in the footer and can be checked with one shell command) and blank forms, without access to the first author’s codes or justifications, on all 120 cells of Table S1 and all rows of Table S2; disagreements will be adjudicated by discussion and the pre-adjudication agreement reported. The second author had read the submitted manuscript, including its condensed Table 1, before the second coding began. Blindness is therefore complete for the per-cell justifications and for the cells added in revision (the M4a–M4c columns and the passive frame and neural subjective frame rows) and partial for the cells whose condensed codes appeared in the submitted version; agreement is reported for the two subsets separately as well as overall. The distinguishability judgements in Table S4 — which pairs of stated predictions differ in sign, magnitude, locus, timing, distribution or condition, and at what confidence — were made by the first coder alone; they are the basis of the contested class for M8. Every cross-theory pair of stated predictions in M8 and M6 (55 pairs) is entered in a deposited pair register with its shared condition, shared observable and a four-way relation — different, jointly compatible, discriminating, incompatible — and a pair-level confidence; the class is derived from that register.

The second coder receives Table S4 and the register only after returning Table S1 and then codes the relations independently; until then the pair judgements are single-coder. [$\kappa = \dots$, raw agreement ... of 120 cells overall; $\kappa = \dots$ for the submitted-version cells; $\kappa = \dots$ for the cells added in revision; ... disagreements adjudicated by discussion; S2 $\kappa = \dots$ — to be completed when the second coding is finished] Tables S1–S4, Supplementary Protocol S5, the codebook, the inclusion criteria and the scripts that produce every count in this section will be deposited at [repository] under DOI [10.xxxx/PLACEHOLDER — archive assembled; DOI to be inserted on public deposit] and accompany the submission.

7.3 Table 1: theories by measurable quantity

Table 1. Stated predictions of 12 coded accounts against 10 measurable quantities (120 cells). Trial-level accounts are tallied first; the two origin-level accounts form a separate block with their own tally. E = EXPLICIT, positive polarity; E $\emptyset$ = EXPLICIT, null polarity (a stated prediction that the quantity makes no difference, or its exclusion from the theory’s function); i = INTERPRETED, positive; $i\emptyset$ = INTERPRETED, null; $\cdot$ = NOT_LOCATED (no statement found; not a prediction of no effect); n/a = NOT_APPLICABLE. † marks a cell flagged for adjudication (16 cells). Every cell of the M4a–M4c columns and of the HOT and HOSS rows is provisional pending the second coding. The last three rows are outputs of the typology script, not codes: stated predictions in the register, discriminating pairs among the cross-theory pairs of stated predictions entered in the pair register (a dash where the column holds fewer than two stated predictions), and the derived class. Six further register entries are derived by the authors and not counted here: P23 (the supramodular M2 prediction extended to access, Section 9.1), P24 (a feeling accompanying purely autonomic conflict, derived from the interoceptive-inference mechanism, pending its proponents’ view), P32 and P33 (the threshold and its equivalence half, Section 9.1), P34 (the Study 2 valence hypothesis, Protocol S5) and P25 (a load effect of policy count, derived from GNWT with a capacity assumption it has not made). IIT (3.0/4.0) denotes integrated information theory in its 3.0 and 4.0 formulations, coded as one account because their stated predictions over these quantities coincide. Condensed from Table S1.

Account	M1 N policies	M2 In-compat.	M3 Valence	M4a Intero. content	M4b Vis-ceral modul.	M4c Auto-nomic conflict	M5 Org-wide coh.	M6 Dim.phen.	M7 Metarep.precision	M8 Cor-tical sig-nature
<i>Trial-level accounts (n = 10)</i>										
GNWT	$\cdot$	$\cdot$	$i\emptyset$	$\cdot$	$\cdot$	$\cdot$	i	$\cdot$	i	E
IIT (3.0/4.0)	$\cdot$	$\cdot$	$\cdot^\dagger$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	E	$\cdot$	E
Recurrent processing (RPT)	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot$	$\cdot^\dagger$	$\cdot$	$\cdot$	E
Higher-order thought (HOT)	$\cdot$	$\cdot$	$i\emptyset$	i	$\cdot$	i	$\cdot$	$\cdot^\dagger$	E	E

Account	M1 N policies	M2 In-compat.	M3 Valence	M4a In-tero. content	M4b Visceral modul.	M4c Auto-nomic conflict	M5 Org-wide coh.	M6 Dim. phen.	M7 Metarep. precision	M8 Cor-tical sig-nature
Higher-order state space (HOSS)	.	.	·†	·†	·†	·†	.	E	E	E
Attention schema (AST)	.	.	.	.	.	.	.	<i>i</i>	<i>i</i>	E†
Predictive processing / beast machine (PP)	.	·†	E	E	<i>i</i> †	<i>i</i> †	<i>i</i>	<i>i</i>	E	<i>i</i>
Supra-modular interaction theory (SIT)	E	E	<i>i</i> ∅	<i>i</i>	.	E∅	.	.	.	<i>i</i>
Passive frame theory (PFT)	<i>i</i> ∅	<i>i</i> ∅	<i>i</i> ∅	<i>i</i>	.	E∅	.	.	·†	n/a
Neural subjective frame (NSF)	.	.	<i>i</i>	<i>i</i> ∅	E†	.	<i>i</i>	<i>i</i> †	.	<i>i</i>
Tally, tri-al-level (n = 10)	1 E / 1 <i>i</i> ∅ / 8 ·	1 E / 1 <i>i</i> ∅ / 8 ·	1 E / 1 <i>i</i> / 4 <i>i</i> ∅ / 4 ·	1 E / 3 <i>i</i> / 1 <i>i</i> ∅ / 5 ·	1 E / 1 <i>i</i> / 8 ·	0 E / 2 E∅ / 2 <i>i</i> / 6 ·	0 E / 3 <i>i</i> / 7 ·	2 E / 3 <i>i</i> / 5 ·	3 E / 2 <i>i</i> / 5 ·	6 E / 3 <i>i</i> / 0 · / 1 n/a
<i>Origin-level ac-counts (n = 2)</i>										
Unlim-ited asso-ciative learning (UAL)	.	.	E	<i>i</i>	.	.	<i>i</i>	.	.	n/a
Feinberg & Mallatt (FM)	.	.	E	E†	.	·†	<i>i</i>	.	.	<i>i</i>
Tally, origin-level (n = 2)	0 E / 2 ·	0 E / 2 ·	2 E / 0 ·	1 E / 1 <i>i</i> / 0 ·	0 E / 2 ·	0 E / 2 ·	0 E / 2 <i>i</i> / 0 ·	0 E / 2 ·	0 E / 2 ·	0 E / 1 <i>i</i> / 0 · / 1 n/a
Tally, all rows (n = 12)	1 E / 1 <i>i</i> ∅ / 10 ·	1 E / 1 <i>i</i> ∅ / 10 ·	3 E / 1 <i>i</i> / 4 <i>i</i> ∅ / 4 ·	2 E / 4 <i>i</i> / 1 <i>i</i> ∅ / 5 ·	1 E / 1 <i>i</i> / 10 ·	0 E / 2 E∅ / 2 <i>i</i> / 8 ·	0 E / 5 <i>i</i> / 7 ·	2 E / 3 <i>i</i> / 7 ·	3 E / 2 <i>i</i> / 7 ·	6 E / 4 <i>i</i> / 0 · / 2 n/a
Stated predic-tions in the re-gister (Table S4)	1	1	3	2	1	2	0	2	3	12

Account	M1 N policies	M2 In-compat.	M3 Valence	M4a In-tero. content	M4b Visceral modul.	M4c Auto-nomic conflict	M5 Org-wide coh.	M6 Dim. phen.	M7 Metarep. precision	M8 Cortical signature
Discriminating pairs / cross-theory pairs registered (pair register)	—	—	—	—	—	—	—	0 of 1	—	2 of 54
Derived class	single-occupant	single-occupant	occupied-not-contested	occupied-not-contested	single-occupant	occupied-not-contested	thin	occupied-not-contested	occupied-not-contested	contested

Every count below is of the form k of n coded accounts and was recomputed by script from the deposited matrix. Across the 120 cells the codes are 21 EXPLICIT (19 positive, 2 stated null), 29 INTERPRETED (22 positive, 7 stated null), 68 NOT_LOCATED and 2 NOT_APPLICABLE. The register holds 27 stated predictions, of which 12 fall in the M8 column and 15 in the other nine columns together. The spread across rows is modest; the spread across columns is not, and the columns are the result.

7.4 Reading the columns

The class of each column was derived by the typology script after the first coding was frozen, and we report the classes as they came out, including the three places where they contradict the earlier version of this review.

Contested: M8. The cortical signature of access is stated by 6 of 12 accounts — GNWT, IIT, recurrent processing theory, higher-order thought theory, higher-order state space theory and the attention schema theory — and is INTERPRETED for 4 more; no account is silent on it, and the two NOT_APPLICABLE cells (passive frame theory, unlimited associative learning) decline the level at which the question is posed. The register holds 12 stated predictions here — the six preregistered GNWT and IIT predictions of the Cogitate protocol [1,2], the local-recurrence prediction of recurrent processing theory [75], the dorsolateral prefrontal prediction of higher-order theory [63,116], the temporoparietal prediction of the attention schema theory [117] and the prefrontal awareness-state prediction of higher-order state space theory [82], among others — and the pair register enters all 54 cross-theory pairs of them. Most differ without conflicting: 42 pairs speak to different observables (a locus beside a timing, a decodability claim beside a connectivity claim) and 10 are jointly compatible under a shared observable. Two are discriminating. The GNWT and IIT predictions on temporal profile (P02, P05) name one observable — content-specific activation across a 0.5, 1.0 or 1.5 s stimulus — and predict values that exclude each other, an offset-locked burst against sustained activation with no offset event; this is the pair the Cogitate consortium operationalised and tested [2], and it is the pair on which the column’s class rests (pair confidence 2). The second, recurrent processing theory against higher-order theory on the necessity of dorsolateral prefrontal function for awareness at matched first-order performance (P08, P09), is entered at pair confidence 1 and flagged: it discriminates only if the two sources’ outcome variables — phenomenal content and metacognitive awareness — are one variable. No pair is incompatible outright. The earlier version of this review counted 15 “distinguishable” pairs here; under the four-way relation 7 of those are different, 6 jointly

compatible and 2 discriminating, and the column is contested on the strength of the two, not the fifteen. The column is shared at the level of the quantity and contested at the level of one value, the temporal profile. The earlier version of this review called M8 “shared and convergent” and said that no occupant could be singled out by a measurement of it; the register says otherwise, and so did the experiment. This is the column on which the theories have done their work: 12 of the 27 stated predictions fall here.

Single-occupant: M1 and M2. The number of simultaneously competing policies (M1) and their degree of incompatibility (M2) are each stated by 1 of 12 accounts, supramodular interaction theory, whose PRISM principle makes the simultaneous arrival of incompatible inclinations at the skeletomotor output system the trigger for conscious conflict [38,39]. Incompatibility there has been measured — sustaining incompatible skeletomotor intentions while motionless produces stronger subjective effects than sustaining compatible ones [40]. Two features of the M1 prediction are recorded in the register (P16) and matter for Section 9. It is a threshold, not a count effect: conscious conflict arises when two incompatible inclinations are active at once, and the sources say nothing about more than two — a NOT_LOCATED about the further increase, not a prediction that there is none [39]. And its outcome variable is experienced conflict, not access to a stimulus (P17); the extension to access that Section 9 tests is our derivation, registered as P23. Passive frame theory is INTERPRETED with null polarity on both columns, since the field is said to present its contents whether or not conflict is present [39,43]; predictive processing, INTERPRETED on M2 in the first pass, is NOT_LOCATED after a full-text reading of its source, which treats conflict between affordances without any statement about consciousness. No other account states or entails a prediction over either quantity. An experiment varying incompatibility cannot adjudicate between GNWT, IIT, recurrent processing theory or the attention schema theory, none of which said anything about it; a positive result corroborates the supramodular prediction and leaves the rest untouched, and the informative outcome is the null.

Occupied but not contested: M3 and M7. Valence (M3) is stated by predictive processing at trial level and by both origin-level accounts at lineage level, with the same sign; 5 accounts are INTERPRETED, 4 of them with a constructed null — GNWT, higher-order thought theory, supramodular interaction theory and passive frame theory admit valenced content and state no valence-dependent access prediction — and the IIT cell is flagged, since its statement that affect is not an essential property of experience can be read as a stated null rather than silence. Metarepresentational precision (M7) is stated by higher-order thought theory, higher-order state space theory and predictive processing [63,82,118], all as a dissociation of a metacognitive index from first-order performance, and the register finds no pair that differs. Both columns can corroborate or disconfirm their occupants together and cannot separate them.

Occupied but not contested, formerly called contested: M4a and M4c; single-occupant: M4b. The earlier version of this review named autonomic involvement as the one column carrying opposed stated predictions — supramodular interaction theory’s stated negative against the interoceptive-inference positive — and prized it as the cell an adversarial test should want. Splitting the column into its three questions dissolves the opposition. Access to interoceptive content as content (M4a) is stated by predictive processing, for which visceral channels are constitutive of feeling [48], and by the Feinberg–Mallatt scheme at lineage level [24], a cell flagged because its

explicit interoceptive category is in a source we could not read; 4 accounts are INTERPRETED positive and the neural subjective frame INTERPRETED null, since the frame is proposed as a referential not itself experienced as content [119]. Visceral modulation of access to exteroceptive content (M4b) is stated by one account, the neural subjective frame, whose anchor is that pre-stimulus heartbeat-evoked responses predict detection of a faint grating independently of measured cardiac parameters [120] — a cell coded from the abstract and provisional until the full text is read; predictive processing is INTERPRETED, re-coded from EXPLICIT because the coded sentence reviews evidence rather than stating a prediction of the framework; 10 accounts are NOT_LOCATED. The column is single-occupant. Conscious involvement in purely autonomic conflict (M4c) carries the two stated nulls: supramodular interaction theory predicts, and has measured, that conflict within a smooth-muscle effector produces no experienced conflict [40], and passive frame theory states the same boundary as scope — the conscious field serves the somatic nervous system [39]. Both exclude autonomic conflict from the function of consciousness and neither denies that bodily states are felt; the same text names tooth pain and urges among the most basic conscious contents [39]. Against these nulls stand no stated positives: the interoceptive-inference sources contain no statement about a conflict resolved by autonomic effectors alone [47,48], the prediction that unresolved interoceptive prediction error would yield a feeling is our derivation (P24), and the one passage bearing on it names predictions rather than prediction errors as the likelier vehicle of conscious feeling [48]. The positives on M4a and the nulls on M4c answer different questions, no pair of stated predictions in either column is discriminating, and each is occupied but not contested; M4b, with one stated occupant, is single-occupant. Were P24 endorsed by its proponents, M4c would become the first contested column outside M8; as coded, it is not. What has not changed is the methodological fact: a test of M4a or M4c needs a report-independent measure of access to interoceptive content, and none has been validated — such measures exist in the visual [15,103] and auditory [106] modalities, and cardiac interoceptive accuracy measures require report by construction [118].

Thin, not empty: M5. Organism-wide coherence across cortical and autonomic channels, a joint measure of the state accompanying access, receives 0 EXPLICIT and 5 INTERPRETED cells among 12 accounts, and 7 NOT_LOCATED. M5 is a construct this review proposes, not a quantity inside any operational definition of access in use, and the column records that no coded theory has adopted it: five are compatible with it and none operationalises it as an organism-wide joint measure. GNWT's global availability is organism-wide in rhetoric and cortical in measurement, its dependent variables being ignition and P3b [121]; allostatic accounts identify a large-scale interoceptive system [50] without proposing its coherence with cortical dynamics as an access criterion. The nearest stated prediction is the neural subjective frame: cortical responses to heartbeats are held to be a precondition of conscious perception [119], and pre-stimulus heartbeat-evoked responses do predict visual [120] and somatosensory detection [122,123], as does respiratory phase [124,125]; they have also been used to detect residual consciousness in post-comatose patients [126]. These measures couple cortex to one visceral channel and are, by the original authors' own control analyses, independent of the measured autonomic state [120]. They therefore index cortical monitoring of a visceral signal rather than coherence across cortical and autonomic channels, and the programme itself frames the quantity as necessary but not sufficient for conscious perception [119] and as distinct from attention [127]. We code it INTERPRETED on M5 and EXPLICIT on M4b, and we with-

draw the earlier formulation that nobody operationalises a cortico-visceral quantity as a condition on access: somebody does, single-channel, and the column is thin rather than empty. A coder who read the heartbeat-evoked response as a cortical–autonomic coherence metric would fill 1 of 12 cells and make the column single-occupant, and Section 7.7 carries that sensitivity.

Occupied but not contested: M6. Dimensionality of phenomenal space is stated by IIT, for which the structure of an experience is identical to the unfolded cause–effect structure of its substrate and therefore high-dimensional [74], and by higher-order state space theory, which nests a one-dimensional awareness state above a high-dimensional perceptual space [82]. The earlier version of this review listed the pair as distinguishable on structure and magnitude at confidence 1. The pair register enters it as *different*: the two statements address different variables — the geometry of what is experienced in the one case, the dimensionality of the judgement that marks a state as experienced in the other — and no shared observable is named under which one value would exclude the other. The column is therefore occupied but not contested, which is what the submitted version said; it returns to contested only if the first coder rules, on the sources, that the two claims concern one variable, and that ruling goes to adjudication. Three accounts are INTERPRETED, and two cells of the column are flagged, one in each direction.

The columns that are single-occupant or thin intersect in a cell no study fills: access to valenced or interoceptive content under a manipulated number of incompatible policies, measured across cortical and autonomic channels jointly. Part of that cell is occupied, and by whom is worth stating precisely. Morsella and colleagues established that policy incompatibility has measurable consequences for experience and that smooth-muscle conflict does not [40,128]; Desender and colleagues established the more directly relevant point that the subjective experience of conflict is a separable measurable quantity, dissociable from conflict adaptation — adaptation occurs without conflict awareness, and awareness of conflict has effects of its own [129], which is the closest existing precedent for treating M2 as a predictor of experience rather than of performance; Garfinkel and colleagues, that interoceptive accuracy dissociates from interoceptive awareness [118]. What none has done is vary conflict load orthogonally to salience while measuring access with a report-independent signature, the constraint the no-report literature established as necessary [102], with objections in [108]. Nor can existing data be redescribed to fill the cell: no reanalysis of the Cogitate dataset can recover an effect of policy conflict, since its stimuli — faces, objects, letters and false fonts under target detection, hold conflict load constant by construction [2].

A gap reducible to unavailable methods is a difficulty, not a gap. Most of what the missing measurement requires is routine, and one component is not; the component that is not — a report-independent measure of access to interoceptive content — was named above under M4a and M4c, and it decides which parts of the design in Section 9 are ready to run. Routine: policy conflict is manipulable and its subjective consequences measurable without overt movement [40]; report-independent access measures are established in the visual [15,103] and auditory [106] modalities; calibrated valenced stimuli [130] and standardised cardiac interoceptive measures [118] are off the shelf; and multi-site preregistered adversarial designs are feasible at scale [1]. The absence is explained instead by what the theories ask for. The field’s own coding shows that support for a given theory is predictable from methodological choices alone, irrespective of findings, and that most

studies interpret results post hoc rather than testing a priori predictions [78]. Where no theory names conflict load or organism-wide coherence as a variable, no paradigm is built to vary it.

7.5 An independently generated column set

Three of the ten columns are the first author's own, and the sensitivity check offered in the earlier version of this section — dropping M1, M2 and M5 and confirming that M8 still dominates — tested a proposition nobody was going to contest. The proposition that needs testing is different: whether the emptiness or thinness of a column is a fact about the theories or an artefact of a column set chosen by someone whose theory occupies it. Emptiness relative to a column set of the first author's choosing is not evidence of anything until the column set is independent. The first check we could run uses the dimensional scheme of the ConTraSt database [78], assembled by people with no stake in this argument to organise 412 experiments, and codes the theory rows against its twelve dimensions (Table S3; 120 cells). The S3 coding predates the present scheme: it carries ten rows rather than twelve, with higher-order thought and state space theories still merged and the neural subjective frame absent, and it uses the earlier four codes, which map onto the present ones as described in Section 7.2. The two missing rows are named as a repair in Section 7.7.

The result has two parts. First, along the independent column set every dimension receives at least one stated prediction; none is empty in the sense that M5 is thin in Table 1. Occupancy is uneven in a way that mirrors the corpus: seven of the twelve dimensions — experimental paradigm, measure type, task type, spatial findings, dependent measure, stimulus, and report versus no-report — carry stated predictions from at least four of the eight neuroscientific rows, while the sparse dimensions are the electrophysiological result fields — frequency findings receive one stated prediction (GNWT) and temporal findings three (GNWT, IIT, recurrent processing theory). Per row the ordering also mirrors the corpus: GNWT states predictions on 11 of 12 dimensions, IIT on 10, recurrent processing theory on 7, passive frame theory on none, because it declines the implementation level. The theories are therefore not uniformly underspecified, and the thinness of M5 is not an artefact of general vagueness: the same accounts that are silent on M5 speak on most of the axes the field's own inventory is organised on. Second, none of M1, M2 or M5 corresponds to a ConTraSt field. The check covered every field name and value in the live schema. No field counts concurrent response options; response conflict is not an annotated variable, the conflict paradigms sitting under a categorical task label with no conflict-magnitude field; and no field records cortical–autonomic coherence, although its components exist separately as cortical connectivity tags and cardiac tags, including the heartbeat-evoked potential. By contrast M7 and M8 are directly represented, and M3, M4a–M4c and M6 are partly recoverable through free-text stimulus and measure labels. The quantities this review calls unmeasured are also unrecorded in the largest theory-neutral inventory of the field.

What this does and does not establish should be stated with equal care. It establishes that the Table 1 result for M1, M2 and M5 is not the rediscovery of an existing axis under another name, and that the accounts coded silent on those columns are not silent in general. It does not establish that the quantities are absent from experiments: ConTraSt records what authors reported, and a conflict-magnitude or cortical–autonomic variable could have been manipulated in a coded study without acquiring a field. It cannot say whether M5 is thin because the quantity is hard to measure or because theories find it uninteresting, nor whether M1 and M2 are the right variables. Nor can it say whether

a column is contested: only its temporal and spatial dimensions record the value of a prediction rather than its presence, so the scheme checks occupancy, not disagreement. The coding has its own limits: the published extraction sheet was inaccessible, so the scheme was reconstructed from the database's methods description and live schema; several GNWT and IIT cells rest on the Cogitate preregistration rather than on foundational texts; the YES code admits explicit invariance claims, which inflates three dimensions, and under a strictly differential reading GNWT falls to 9 stated predictions, IIT to 8; and it was done by one coder, with a blank form in the second-coder package. The second independent route — writing to proponents of four or five of the theories and coding against the union of what they name — has not been taken, and we name it as the next check rather than claim its result.

7.6 Table 2: content inventory

The inventory asks a different question of the same theories: what content have their tests actually manipulated? Its unit is now the experiment rather than the publication: a publication reporting several experiments contributes several rows, control and ancillary experiments are rows flagged as such, and where only the abstract could be read the publication is carried as one row with the limitation recorded. The sample is purposive — the primary statements of the coded theories, the studies those statements cite as support, and the adversarial-test literature, under the deposited inclusion criteria: 42 experiments from 32 publications. Four further publications — the Cogitate protocol, the taxonomy paper of the masking family, the front-versus-back review and the theory paper behind the sustained-intentions paradigm [1,121] — define paradigms rather than report data; they were rows of the earlier inventory, are now listed as paradigm sources, and are outside every denominator. Content is coded on two orthogonal fields, modality and affective status, from which the class is derived by script; interoceptive content means content from visceral afferents or an autonomic signal modulating access, and motor conflict is split by effector, skeletal or autonomic. The category that does the analytic work is “theory-addressed”, and its rule (criterion I2) keeps three attributions apart: a study is theory-addressed when its own authors name a coded theory as tested or as the framework their result bears on, or when a later paper that is a coded theory's primary statement cites it; the coder's own inference is recorded separately and enters only a sensitivity scenario. Under that rule the baseline denominator is 30 experiments from 21 publications, with one publication whose access was suppressed rather than manipulated excluded [131], the five metacognition experiments included as contested rows, and one interoceptive attention study included but contested [132].

Table 2. Content whose access was manipulated in the experiments the coded theories are tested by or built on (Table S2). Counts are experiments, as k of n ; the third column is the baseline scenario S0 of the deposited sensitivity file and is the denominator used in the text. Of the 42 experiments, the two state experiments without a content contrast and the one experiment in which access was suppressed by masking rather than manipulated [131] are outside both columns; the latter enters only as a sensitivity scenario below.

Content class	All content experiments, no theory filter ($n = 39$)	Theory-addressed, baseline S0 ($n = 30$)
Neutral visual	22 of 39	18 of 30

Content class	All content experiments, no theory filter (n = 39)	Theory-addressed, baseline S0 (n = 30)
Competing motor policies, skeletal effectors	7 of 39	7 of 30
Competing motor policies, autonomic effector	1 of 39	1 of 30
Neutral non-visual	4 of 39	3 of 30
Valenced	3 of 39	0 of 30
Interoceptive (visceral afferent content)	2 of 39	1 of 30

The finding is an imbalance, and it should be stated in the terms the attribution rule licenses. In the checked sample, under the stated attribution rule, 18 of 30 theory-addressed experiments manipulated neutral visual content, 8 of 30 competing motor policies — 7 realised by skeletal effectors and 1 by an autonomic effector, all from the supramodular programme — 3 of 30 neutral non-visual content (the auditory no-report family [104]), 0 of 30 valenced content and 1 of 30 interoceptive content. The single interoceptive experiment is the cardiac interoceptive-accuracy study that a coded theory’s primary statement cites [118], itself a contested inclusion; in the strict scenarios the cell is empty. The smooth-muscle control experiment of the supramodular programme [40] was coded interoceptive in the earlier version; it is re-coded here as autonomic-effector motor conflict, because the content whose access it measures is the experienced conflict between a sustained intention and the pupillary reflex, not a visceral afferent signal. It is the only theory-addressed experiment on autonomic-effector conflict in the sample — a control condition inside one programme — and it is the empirical anchor of column M4c. The valenced cell is empty because the three valenced experiments in the sample — fearful expressions breaking continuous flash suppression faster than neutral ones [130], arousal and valence varied orthogonally [133], threat-signalling input gaining preferential access [134] — are named by no coded theory’s own authors and cited by no coded theory’s primary statement. That is a statement about this sample and this rule, and we withdraw the earlier version’s formulation that no coded theory claims the valenced or interoceptive literature: a registered search, or a broader reading of what counts as a theory’s primary statement, could attach a theory to any of these studies.

How sensitive the counts are is stated in full, from the deposited scenario file. Adding the masked-expression study whose access was suppressed [131] makes valenced content 1 of 31. Removing the five metacognition experiments gives neutral visual 14 of 25. Accepting a later attribution only when its citing sentence was read and names the theory gives 13 of 24; removing every contested row as well gives 10 of 21, the smallest neutral-visual share in any scenario. Adding the coder’s own inferred attributions gives 22 of 35. Dropping the theory filter gives 22 of 39 neutral visual, 3 of 39 valenced and 2 of 39 interoceptive. Seven rows carry a contested-inclusion flag with a written reason. The publication-level counts of the earlier version reproduce from the legacy codes — 21 of 27 with the paradigm-defining rows, 19 of 24 without — so the change of unit and of attribution rule, not recoding, accounts for the difference between versions. The imbalance survives every scenario; the zero in the valenced cell survives every theory-filtered scenario except the one that admits the suppressed-access study; and the abstract states the zero together with the neutral-visual count as one imbalance, under the attribution rule named in this section. The pattern is corroborated from outside our own coding: the field’s census of 412 experiments codes stimulus modality and category as methodological variables and reports no valence breakdown at all [78]. One limitation belongs

here rather than in Section 7.7: 19 of the 32 publications could be read only as abstracts, so for those the partition into experiments and the participant counts are provisional.

7.7 What survives reclassification, and limitations

Because the INTERPRETED/NOT_LOCATED boundary carries the weight, we tested the derived classes against the least favourable reclassification available: promoting every INTERPRETED cell to EXPLICIT, that is, deciding every judgement call against our own conclusion. The M1 and M2 results survive in the sense that matters for Section 9. M1 has 1 EXPLICIT and 1 INTERPRETED cell, and the INTERPRETED cell is passive frame theory's constructed null; M2 has 1 EXPLICIT and 2 INTERPRETED, passive frame theory's null and predictive processing's flagged positive. Promotion would place a stated supramodular positive beside a stated passive-frame null — a pair the typology script would count as differing in sign, making the column contested between two formulations of one programme whose difference Section 4 describes. We therefore do not claim that no reading makes M1 or M2 contested; we claim that as coded they are single-occupant, and that the only reading that changes this promotes a null the passive-frame authors have not stated. The M5 result does not survive: promoting its five INTERPRETED cells turns a column with no stated prediction into 5 of 12, and a coder who read the heartbeat-evoked response as a joint cortical-autonomic measure would fill one cell even without promotion. The claim that M5 is thin therefore rests on the judgement that a rhetorical organism-wide commitment without an operationalisation, and a single-channel cortico-visceral predictor, are not stated predictions over organism-wide coherence. We think that judgement is right and have given the per-cell reasons in Section 7.4, but it is a judgement, which is why M5 is a secondary and qualified claim in this review. The M8 class is unaffected by promotion, because it rests on 6 EXPLICIT cells and on the discriminating GNWT-IIT pair on temporal profile, and the M4a–M4c classes are unaffected in the other direction: promotion adds occupants of the same sign and creates no discriminating pair. The M6 class turns on whether the IIT and higher-order state space statements concern one variable; the pair register says they do not, and the column is occupied but not contested unless adjudication rules otherwise.

A second sensitivity runs the other way. Three of the 21 EXPLICIT cells carry an adjudication flag: the attention schema theory on M8, whose locus claim is not in the source the earlier version cited [84] but in an earlier statement of the theory [117]; the Feinberg–Mallatt scheme on M4a, whose explicit interoceptive category is in a book we could not read; and the neural subjective frame on M4b, whose stated prediction is coded from an abstract we could not get behind. Downgrading the first leaves M8 contested with 5 EXPLICIT cells. Downgrading either of the other two leaves its column with a single EXPLICIT cell and makes it single-occupant rather than occupied but not contested; neither downgrade creates a contested column. The 15 flagged cells are listed with their reasons in Table S1 and are the first business of the adjudication meeting. The sensitivity analysis is prepared for deposit with the coding.

Five limitations bear on how much weight Tables 1 and 2 can carry, and we state them without mitigation. First, the codes reported here are those of a single coder; the blind second coding is described in Section 7.2 and its agreement statistics are reported there once complete. Until then the evidential base of this section is 120 cells, 31 register entries and 42 experiment classifications produced by one reader applying a written scheme to texts, and we make no reliability claim. Second,

the evidence base is partly second-hand: 58 of the 120 cells, and 28 of the 54 EXPLICIT and INTERPRETED cells, rest on abstracts rather than full text, as do the locus claims of recurrent processing theory, higher-order thought theory and the attention schema theory in the register [63,75,76]; the body text of the founding supramodular statement [38] could not be read, so P16 is verified only from the later synthesis [39]; and 19 of the 32 inventory publications were read as abstracts. Each such cell is marked in the deposited tables, and a stated prediction resting on an abstract is a weaker warrant than one resting on a quoted passage. Third, both the theory set and the study set are purposive samples; the accounts that meet the inclusion rule and were not coded are named in Section 7.1, and a registered search would very likely change the counts in Table 2, substantially so for the smaller cells. Fourth, the column set is the first author's, and the only independence check performed is the ConTraSt coding of Section 7.5, which covers ten of the twelve rows and checks occupancy rather than disagreement; the letter-to-proponents route, which would also settle whether P24 is a prediction its proponents own, has not been taken. Fifth, coding a "predictive apparatus" requires deciding which formulation of a theory counts as canonical, and for theories with several revisions we coded the most recent statement making empirical commitments — defensible but not neutral, and the reason two pairs of rows that were one are now two.

8. Online metarepresentation, policy candidacy and reconstruction

The upper part of the episode — registering one's own processing, treating a proposed action as one option among several, reconstructing it afterwards — holds three dissociations that are framed but, with the partial exception conceded in Section 8.2, not measured, each conflated with a measured neighbour.

8.1 Precision, not registration

The case for treating self-registration as a separate quantity is not introspective: the accuracy of judgements about one's own processing varies separately from that processing's accuracy. Second-order signal detection supplies the apparatus, meta- d' expressing metacognitive sensitivity in units of first-order d' [135]; calibration, error sensitivity and performance dissociate, so "metacognition" unqualified means little [136]. Whether theta-burst TMS to dorsolateral prefrontal cortex impairs metacognitive sensitivity with discrimination intact is disputed. Rounis and colleagues reported the impairment [116]; a replication with a modified design found none and pointed to methodological differences from the original [137]; Ruby and colleagues argued that the replication's own pre-exclusion data showed the effect and that a Bayesian re-analysis favours it [138]; Bor and colleagues replied that the prior of that re-analysis rests on data they consider invalid and that the evidence across studies does not yet demonstrate the effect [139]. Neither side has conceded, and we decide nothing here; the higher-order locus prediction in the register (Table S4, P09) carries the dispute as an auxiliary assumption. Lesion data implicate anterior prefrontal cortex, whose integrity predicts introspective rather than perceptual accuracy [140,141], with domain-general and domain-specific components [142] and a separate evaluative stage [143]. Registration itself grants no control. But its *precision* predicts downstream change — confidence governs information search [144], poor metacognition accompanies resistance to counter-evidence [145], symptom dimensions shift metacognitive bias without touching performance [146]. The measurable variable is metarepresentational precision, not the fact of registration — a different claim from the one most theories in Table 1 make.

Two qualifications. Self-registration may not be self-directed: confidence compares contributions to joint decisions [147], is arguably a culturally acquired skill over implicit monitoring [148], secondary to mind-reading [149], and what a workspace needs to select what is admitted [150]; the radical version makes consciousness a learned metarepresentation [151–153]. And these claims rest on non-equivalent report scales: the Perceptual Awareness Scale exists because binary seen/unseen flattens intermediate states [154] and is more sensitive to weak contents than confidence [155], the discreteness dispute is unresolved [156] against workspace theories predicting a threshold [157], and scale and quantity are poorly coordinated [158].

8.2 Policy candidacy

The second dissociation concerns **policy candidacy** (polyvariant choice) — the mode in which a proposed action policy ceases to be the sole command and becomes one member of a compared set. Three features define it: a change in the proposal’s status, a comparison set, and independence from both online metarepresentation and the success of control.

The name requires justification, because the obvious label is taken. In computational neuroscience *meta-control* means arbitration between controllers and tuning of their parameters: model-based versus model-free learning by reliability [159], deciding how to decide [160], allocating effort by expected benefit [161], reviewed in [162]. That is parametric arbitration, not a policy represented as an object; since the term denotes the very neighbour this mode must be distinguished from, we do not use it, and this is its only appearance.

Existing paradigms measure everything adjacent to policy candidacy and not it. Meta-d’ measures the accuracy of evaluating a decision already taken [135]. The expected value of control measures the *intensity* of allocation in dorsal anterior cingulate cortex [163,164] over a cascaded prefrontal organisation [165] whose capacity limit is probably task separation, not fuel depletion [166] — the resource model having failed a preregistered replication [167] and multi-site test [168], effort recast as opportunity cost [169]. The stop-signal paradigm measures the speed of cancelling one prepared action [170,171]; its unconscious variant is decisive, masked stop signals recruiting frontal inhibitory machinery without awareness [172,173], conscious and unconscious control engaging dissociable rather than graded mechanisms [174]. Conflict detection likewise has a canonical account in which the anterior cingulate recruits control with no commitment to consciousness [175]. The closest construct is intentional inhibition, where refusal is itself a choice [176,177] — but there too no alternative is generated and compared.

A correction here strengthens the case. The first of the three quantities operationalising policy candidacy — the size of the set under consideration — *is* already measured by option generation [178,179], including its neural basis [180], so the claim that none is measurable is too strong. **The second quantity — whether a decision depends on properties of the *rejected* alternatives — has an existence proof but no measurement of the kind this section needs. The decoy effect shows that adding an asymmetrically dominated alternative, which nobody chooses, changes the choice between the two that remain [181]; the dependence therefore exists. What the decoy literature does not address, and was not designed to address, is whether the rejected alternative was accessed, whether the shift in choice depends on that access, or whether anything about the rejected option is represented as a candidate rather than as a feature of the display. The paper that established the effect says nothing**

about consciousness, and we cite it for the existence of the dependence only. Unmeasured, then, are the access-dependence of that effect and whether metacognitive precision predicts option-set widening rather than correctness. The metric for the first, **option-set widening**, should be borrowed from option generation, not reinvented; the nearest precedent for the second is that the selection threshold shifts with urgency [182].

8.3 Report as write-back into the self-model

The third dissociation concerns what report is evidence for. That participants report not the causes of their behaviour but plausible theories of those causes remains established, if localised more precisely than originally [183], extending to introspection about current phenomenal content [184]. Choice blindness is the strong version: when already-chosen faces were switched, many switches went unnoticed and participants justified a choice they had not made [185], reproduced outside the laboratory [186].

The result that matters most is stronger than the standard reconstruction claim: when two of their own ratings on a moral survey were reversed by a paper-slip trick, most participants failed to detect the reversal, and about half argued for the position opposite to the one they had given moments earlier [187]. The study measured immediate acceptance and justification within the session, not the durability of the altered attitude, which its authors raise only as a possibility; the write-back claim below rests on the immediate result. Reconstruction is therefore not a distortion of the record but a write-back into the model of oneself — report can become a cause of the subsequent state. The same holds for the sense of agency, an inference rather than access to the motor command [188,189], and for the construction of having chosen freely [190]. Two cautions: intentional binding [191] is contested, reproduced under illusory voluntary action [192] and sensitive to phenomenological control [193]; and the Libet result must be stated exactly — the averaged readiness potential [194] is not refuted as a measurement, its causal interpretation is, since stochastic accumulation makes it an artefact of averaging [195,196]. Hence “post-hoc reconstruction” rather than “reflection”: the latter imports a transparency the data do not support, itself analysed as the unavailability of the model’s own modelhood to the system using it [197–199].

8.4 What is measured and what is not

The three dissociations do not stand in the same place, and the ledger is uneven rather than empty. Metarepresentational precision has validated instruments — meta- d' and its calibration variants — and no experiment that asks whether precision, rather than registration, gates access; that test can be run with what exists. Policy candidacy is measured in part: option generation already counts the set under consideration (Section 8.2), and the decoy effect shows that a decision can depend on properties of a rejected alternative, whereas whether that dependence runs through access to the rejected option has no measurement at all. Write-back has been demonstrated [187] but has not been turned into a measure that any theory is committed to. What the three share is therefore not absence of measurement but absence of a claimant: none is the dependent variable any theory in Table 1 predicts, and Section 7 documents what that costs. Each would be cheaper to test than the localisation dispute that has absorbed the field’s resources; whether any of them is tested depends on whether some theory is prepared to put a prediction on it.

9. Conclusion: the minimal measurement

The field's problem is not a shortage of theories or of rigour. It is that the coded theories concentrate their stated predictions on one of ten measurable quantities — the cortical signature of access, where one pair of stated predictions, on temporal profile, is discriminating and where the first adversarial test accordingly separated its two targets — and have stated almost nothing about the other nine. Of the 27 stated predictions in the register, 12 fall in that column. The rest of the space is not contested; it is, for the most part, unspoken for. That is what Cogitate demonstrated, and the consortium said so: the challenges extend to theories sharing the predictions tested [2]. The derived classes of Section 7.4 say what each of the other columns can carry. The number and incompatibility of competing action policies are single-occupant: supramodular interaction theory states a threshold over the first and an effect over the second, confines both to skeletal muscle by construction [38,39], and no other coded account states a prediction over either. Valence and metarepresentational precision are occupied by accounts that agree. The three autonomic columns are occupied by accounts answering different questions, and no pair of stated predictions differs. Organism-wide cortical-autonomic coherence is stated by no coded theory and is compatible with five. And in the checked sample of theory-addressed experiments, under the stated attribution rule, valenced content is absent.

One experiment converts part of that description into a result, and we state exactly which part. Hold salience, task relevance and the number of distinguishable responses fixed. Vary how many of the responses the content demands share an effector and so cannot be executed together. In a separate branch, vary how many responses are demanded at all, with none sharing an effector. Measure access with a report-independent signature, and record autonomic channels alongside it.

The design replaces the one in the earlier version of this review, and the reason is a confound the earlier design carried. There, incompatibility was set by the target's feature conjunction: three rules demanded three identical keys, two identical and one different, or three different keys. The number of distinct responses therefore rose with incompatibility, so that the incompatibility contrast was also a contrast in response-set size and in congruency, and any effect could have been attributed to either. The supramodular sources themselves locate the gradation of conscious conflict along the compatibility axis and not the number axis — compatible plans may co-occur without any perturbation of consciousness [39] — so the earlier design could not have tested what the theory states. In the present design the target always demands three distinct responses, one per rule, and only the effector mapping in force for the block decides whether they can be executed jointly. Number of policies is varied only in the separate branch at zero incompatibility, where a load account — GNWT joined to a capacity assumption it has not made, registered as the derived P25 — would predict a decrement.

9.1 One worked condition: incompatibility at fixed count under no-report

What is tested, and by whom it was predicted. Study 1 tests two author-derived operationalisations (P32, P23) of one account's stated commitments and estimates a third quantity. Supramodular interaction theory states P16: conscious conflict arises when two or more incompatible inclinations toward the skeletomotor output system are active at once; nothing is stated beyond two [38,39]. The count branch does not test P16: it varies co-executable, not incompatible, policies and measures access to a masked target, not experienced conflict. P32 is therefore ours: access is higher with two or more co-executable policies than with one. P33, also ours, holds that access does not rise further from

two to three — the source is silent beyond two, and silence is not a stated null — and is bounded by an equivalence test, not tested. P23 is ours as well: access rises with the incompatibility of the licensed policies at fixed count and load; the theory's stated outcome is experienced conflict [40]. None is endorsed by the theory's proponents; a null constrains our derivation, not the theory. The condition tests no stated prediction of any theory directly and discriminates between no two theories.

Design. Three overlearned rules map a masked target's hue, position and aperture shape to three mutually distinct responses. Responses are positions of three three-position levers (each hand, dominant foot); two responses can be executed together only on different levers. Incompatibility is the number of pairs of demanded responses sharing a lever — 0, 1 or 3 — set by the effector mapping in force for the block. Effector balancing gives 11 presentation cells and 5 analysis cells: the count branch, counts 1, 2 and 3 at 0 shared pairs, and count 3 at 1 and at 3 shared pairs. Cells are blocked, not interleaved: 11 cells $\times$ 2 blocks per session, mapping fixed per block and cued at its start, order counterbalanced. On a quarter of trials a probe names one rule and the participant executes that rule alone; joint execution is never required. Whether incompatible intentions form at target onset or rules are merely held until the probe is open; probe-response latency by shared-pairs level is the manipulation check bearing on it, reported not tested.

Trial budget. Session 1 supplies titration, a Perceptual Awareness Scale rating on every trial [154] and classifier training: 64 presented trials per presentation cell (a design choice), 704 in all, about 115 min including breaks. Session 2 supplies the confirmatory no-report data without the rating: 104 presented trials per presentation cell (26 probe, 78 no-probe), 1144 in all, about 135 min at 6 s per trial with breaks, giving $78 \times 0.85 = 66.3$ expected usable no-report trials per cell at an assumed 15 % loss. $N = 120$ (119 required for 90 % power at $d_z = 0.30$ for one uncorrected contrast, rounded up to a multiple of the six rule-to-effector rotations), fixed in advance, no sequential monitoring (Supplementary Protocol S5).

Access index. Trial-wise decoded evidence for “seen” from a per-participant classifier trained on the rating-labelled session-1 trials: regularised linear discriminant analysis on 100–500 ms post-target amplitudes at 17 parieto-occipital channels of a 64-channel montage, rating binarised as 1 against 2–4, nested cross-validation with block-wise folds, specification to be preregistered and frozen before session 2 is scored [103,200–202]. The index is EEG-only. It is a candidate marker of access, not a measure of M5; because a no-report participant may covertly categorise [102,108], every contrast is also computed on the session-1 rating.

Autonomic recording. ECG, skin conductance and pupil are recorded throughout, with a jittered 2.5–3.5 s inter-trial interval and deconvolution of overlapping responses. A cortical–autonomic coupling index (the per-participant slope of decoded access on pre-stimulus heartbeat-evoked amplitude; its trial-wise correlation with the phasic autonomic responses) is reported with estimate and interval per cell as estimation only; no coded theory predicts over it.

Contrasts, multiplicity and power. Two confirmatory contrasts, Holm-corrected at family-wise $\alpha = 0.05$: C0 tests P32, count 1 against the mean of counts 2 and 3 (weights $-1, \frac{1}{2}, \frac{1}{2}$) at 0 shared pairs; C1a tests P23, 0 shared pairs against the mean of 1 and 3 at count 3. An equivalence test of count 3 against 2 bounds P33 within ± 0.30 d_z by two one-sided tests [203]; C1b, a trend linear in shared pairs and a two-degree-of-freedom omnibus, is exploratory. A contrast counts as supported only if its EEG-

index and PAS-report versions are both significant after correction, with the same sign. Each confirmatory contrast singly has uncorrected power 0.90 at $N = 120$ and $dz = 0.30$; at least 0.84 after Holm correction (the Bonferroni level; 0.89 if the two p-values were independent); for the conjunctive rule the product under independence is 0.71 to 0.80, to be revised from the pilot correlation between versions. Equivalence power within $\pm 0.30 dz$ at $N = 120$ is 0.895.

Mandatory pilot. Three criteria, thresholds to be preregistered in S5, must be met before the confirmatory sample runs: the masking staircase holds visibility inside its target band; the classifier, trained at one incompatibility level, reaches $AUC \geq 0.70$ at each other level — a validity test defining the population to which the result applies (awareness decodable at that accuracy) without selecting for the hypothesised effect; and full-contrast accuracy on the single probed rule reaches criterion in every cell. Failure on any criterion stops the study; the pilot is reported regardless.

Valence. Study 2 repeats the design with negative, neutral and positive content, testing negative against positive at matched arousal. This tests P34, our derivation, not the stated P13, which concerns experienced valence and free-energy change, not sad against happy access. Pupil and skin conductance are manipulation checks for arousal matching and enter only a sensitivity analysis as covariates, being possible consequences of the manipulation. The incompatibility $\times$ valence interaction is estimated, not tested, for want of power (S5). A valence effect would corroborate our derivation and separate none of the coded theories.

9.2 The condition that cannot yet be run, and what would make it adversarial

The earlier version of this review presented an autonomic-only condition — a conflict resolved wholly by autonomic effectors, with posture fixed and respiratory pattern change blocked — as the most discriminating test in principle, because it would set supramodular interaction theory's stated null against an interoceptive-inference positive. Section 7.4 has withdrawn the second half of that opposition: the interoceptive-inference sources state nothing about autonomic-only conflict, and the positive is our derivation (P24). The condition therefore tests, as things stand, two stated nulls and one derivation, and becomes adversarial only if proponents of interoceptive inference endorse P24 or state a prediction of their own. We have named the letter-to-proponents route in Section 7.7 as the next check; this is the cell it would most directly affect.

The design difficulty is unchanged and we flag it rather than leave it for a reviewer. Most homeostatic crises recruit skeletal effectors — respiration through the diaphragm and intercostals, thermoregulation through shivering and posture, so breath-holding is not evidence against a skeletomotor boundary. What is required is a conflict whose resolution is wholly autonomic: antagonism between thermoregulatory and gastrointestinal demands with posture fixed and respiratory pattern change blocked, or orthostatic and thermal loading of the baroreflex without postural correction. Deafferentation (high spinal injury, locked-in syndrome) is not a discriminator either, since the claim concerns intentions rather than execution and the central skeletomotor plan is preserved. Blocking respiratory pattern change is itself a skeletomotor intervention and does not escape the objection: in an awake participant, skeletomotor intentions survive any fixation of posture or breathing, and supramodular interaction theory is a claim about intentions.

The deeper obstacle is the one named in Section 7.4: the condition requires a report-independent measure of access to interoceptive content — a signature tracking, trial by trial, whether an

interoceptive event was accessed, validated against report in a calibration set, generalising to no-report blocks, and dissociable from afferent strength, arousal and attention to the body, which alone modulates the heartbeat-evoked potential [204,205] — and no such measure has been validated. The order of work is therefore the reverse of the order of interest. The condition of Section 9.1 uses established visual no-report measures and can be run now; it tests two author-derived operationalisations (P32, P23) of one account's stated commitments and estimates a third quantity, and its informative outcome for the derivations is a bounded null. The autonomic condition would need both a validation study and a stated prediction from the other side before it could adjudicate anything. That is the actual conclusion of this review, and it is a smaller claim than an adversarial test on a new quantity — but it names both the measurement and the statement whose absence keeps the cell closed. Presenting the autonomic condition as ready would be the kind of overclaim this review has been criticising.

A closing note on what this review has and has not done. We have not proposed a theory. We have documented, by coding, that the field's stated predictions are concentrated on one quantity and that most of the remaining quantities are single-occupant, thin or occupied by accounts that do not disagree, and specified the experiment built on the one stated prediction the audit found outside that quantity, which tests two author-derived operationalisations of it rather than the prediction itself. A theory that does make predictions about the single-occupant and thin coordinates — about the format in which incommensurable local evaluations become comparable, its dimensionality as a function of the number and heterogeneity of conflicting domains, and organism-wide reconfiguration breadth as a second condition on access, exists and is set out separately, as Arbitration Format Theory, in a companion paper. It does not propose an alternative physical basis for consciousness, and its claims are stated over observables rather than over experience. Whether it is right is a separate question from whether the columns are single-occupant or thin. They are so either way, and the experiment that operationalises the one stated prediction among them is worth running whoever turns out to have predicted the answer.

Supplementary Materials

Supplementary Materials: The following will be deposited at [repository] under DOI [10.xxxx/PLACEHOLDER — archive assembled; DOI to be inserted on public deposit] and accompany the submission. Table S1: theory-by-quantity coding matrix (120 cells: 12 coded accounts by 10 quantities, with code, polarity, relation, per-cell justification, source DOI and locus, evidence status, confidence, provisional and adjudication flags, and the legacy code of the earlier version where one exists). Table S2: content inventory (42 experiments from 32 publications, with paradigm, modality, affective status, derived content class, the three attribution columns and the derived theory-addressed flag, contested-inclusion flags with reasons, and evidence status), with `paradigm_sources.csv` (the four paradigm-defining publications and the rows that rely on them) and `s2_sensitivity.csv` (every scenario reported in Section 7.6). Table S3: coding of ten theory rows against the twelve ConTraSt annotation dimensions (120 cells). Table S4: register of 34 entries (27 stated predictions, 1 kept for provenance after re-coding to INTERPRETED, 6 derived by the authors and labelled so) with source, locus, condition, outcome measure, expected result and confidence; `pair_register.csv`: every cross-theory pair of stated predictions in M8 and M6 (55 pairs) with shared condition, shared

observable, four-way relation, favouring results and pair-level confidence. Supplementary Protocol S5: full protocol of Study 1 and Study 2 (stimuli, task, effector mapping, cell list, presented and usable trials, exclusions, index freezing, autonomic channels and coupling metric, contrasts and weights, power tables, pilot criteria). Codebook v2 (frozen; version and hash printed in its footer), the inclusion criteria for Tables S1 and S2, the blank second-coder forms, and the scripts that produce every count in Section 7 and every power figure in Section 9 (`column_typology.py`, `agreement.py`, `section9_power.py`, `reproduce.py`).

Data Availability Statement

Data Availability Statement: Tables S1–S4, Supplementary Protocol S5, the codebook, the inclusion criteria and the scripts will be deposited at [repository] under DOI [10.xxxx/PLACEHOLDER]; the archive is assembled and the DOI will be inserted on public deposit. No new empirical data were generated.

Declarations

Author Contributions: Conceptualisation, [Author 1]; methodology, [Author 1] and [Author 2]; investigation (first coding of Tables S1–S4 and of the content inventory), [Author 1]; validation (blind second coding of Tables S1 and S2 against the frozen codebook — scheduled; its agreement statistics will be inserted in Section 7.2 when complete), [Author 2]; software (typology, agreement and power scripts), [Author 1]; writing – original draft, [Author 1]; writing – review and editing, [Author 1] and [Author 2]. Both authors have read and agreed to the published version of the manuscript.

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Use of Generative AI: The authors used generative AI tools [name and version to be inserted by the authors] under their direction for literature retrieval, for checking the coding tables and the power analysis against source records and script output, and for drafting and editing the text. All cited sources were verified by the authors against CrossRef and DataCite records, every count in Sections 7 and 9 was recomputed by the deposited scripts, and the authors take full responsibility for the content of the manuscript.

Conflicts of Interest: The authors declare no conflicts of interest. [Authors: the manuscript argues at length with supramodular interaction theory and names a companion theory of the first author's; any professional relationship with the proponents of the coded theories, and the relationship to the companion paper, belong here.]

References

1. Melloni, L.; Mudrik, L.; Pitts, M.; Bendtz, K.; Ferrante, O.; Gorska, U.; Hirschhorn, R.; Khalaf, A.; Kozma, C.; Lepauvre, A.; et al. An adversarial collaboration protocol for testing contrasting predictions of global neuronal workspace and integrated information theory. *PLOS ONE* **2023**, *18*, e0268577. <https://doi.org/10.1371/journal.pone.0268577>
2. Cogitate Consortium; Ferrante, O.; Gorska-Klimowska, U.; Henin, S.; Hirschhorn, R.; Khalaf, A.; Lepauvre, A.; Liu, L.; Richter, D.; Vidal, Y.; et al. Adversarial testing of global neuronal workspace and integrated information theories of consciousness. *Nature* **2025**, *642*, 133–142. <https://doi.org/10.1038/s41586-025-08888-1>
3. Naccache, L.; Sergent, C.; Dehaene, S.; Wang, X.J.; Farisco, M.; Changeux, J.P. GNW theoretical framework and the “adversarial testing of global neuronal workspace and integrated information theories of consciousness”. *Neuroscience of Consciousness* **2025**, *2025*, niaf037. <https://doi.org/10.1093/nc/niaf037>
4. Baars, B.J. Global workspace theory of consciousness: toward a cognitive neuroscience of human experience. In *Progress in Brain Research*; Elsevier, 2005; pp. 45–53. [https://doi.org/10.1016/s0079-6123\(05\)50004-9](https://doi.org/10.1016/s0079-6123(05)50004-9)
5. Block, N. On a confusion about a function of consciousness. *Behavioral and Brain Sciences* **1995**, *18*, 227–247. <https://doi.org/10.1017/s0140525x00038188>
6. Block, N. Perceptual consciousness overflows cognitive access. *Trends in Cognitive Sciences* **2011**, *15*, 567–575. <https://doi.org/10.1016/j.tics.2011.11.001>
7. Block, N. *The Border Between Seeing and Thinking*; Oxford University Press: New York, NY, USA, 2023. <https://doi.org/10.1093/oso/9780197622223.001.0001>
8. Aru, J.; Bachmann, T.; Singer, W.; Melloni, L. Distilling the neural correlates of consciousness. *Neuroscience & Biobehavioral Reviews* **2012**, *36*, 737–746. <https://doi.org/10.1016/j.neubiorev.2011.12.003>
9. Aru, J.; Suzuki, M.; Rutiku, R.; Larkum, M.E.; Bachmann, T. Coupling the State and Contents of Consciousness. *Frontiers in Systems Neuroscience* **2019**, *13*, 43. <https://doi.org/10.3389/fnsys.2019.00043>
10. Mudrik, L.; Faivre, N.; Pitts, M.; Schurger, A. On a confusion about there being two types of consciousness. *Trends in Cognitive Sciences* **2026**, *30*, 687–699. <https://doi.org/10.1016/j.tics.2025.11.012>
11. Birch, J.; Schnell, A.K.; Clayton, N.S. Dimensions of Animal Consciousness. *Trends in Cognitive Sciences* **2020**, *24*, 789–801. <https://doi.org/10.1016/j.tics.2020.07.007>
12. Birch, J. The search for invertebrate consciousness. *Noûs* **2022**, *56*, 133–153. <https://doi.org/10.1111/nous.12351>
13. Bayne, T.; Seth, A.K.; Massimini, M.; Shepherd, J.; Cleeremans, A.; Fleming, S.M.; Malach, R.; Mattingley, J.B.; Menon, D.K.; Owen, A.M.; et al. Tests for consciousness in humans and beyond. *Trends in Cognitive Sciences* **2024**, *28*, 454–466. <https://doi.org/10.1016/j.tics.2024.01.010>
14. Mudrik, L.; Faivre, N.; Koch, C. Information integration without awareness. *Trends in Cognitive Sciences* **2014**, *18*, 488–496. <https://doi.org/10.1016/j.tics.2014.04.009>
15. Frässle, S.; Sommer, J.; Jansen, A.; Naber, M.; Einhäuser, W. Binocular Rivalry: Frontal Activity Relates to Introspection and Action But Not to Perception. *The Journal of Neuroscience* **2014**, *34*, 1738–1747. <https://doi.org/10.1523/jneurosci.4403-13.2014>
16. Cohen, M.A.; Dennett, D.C. Consciousness cannot be separated from function. *Trends in Cognitive Sciences* **2011**, *15*, 358–364. <https://doi.org/10.1016/j.tics.2011.06.008>
17. Kouider, S.; de Gardelle, V.; Sackur, J.; Dupoux, E. How rich is consciousness? The partial awareness hypothesis. *Trends in Cognitive Sciences* **2010**, *14*, 301–307. <https://doi.org/10.1016/j.tics.2010.04.006>
18. Maturana, H.R.; Varela, F.J. *Autopoiesis and Cognition: The Realization of the Living*; Boston Studies in the Philosophy of Science, Vol. 42; D. Reidel: Dordrecht, The Netherlands, 1980.
19. Di Paolo, E.A. Autopoiesis, Adaptivity, Teleology, Agency. *Phenomenology and the Cognitive Sciences* **2005**, *4*, 429–452. <https://doi.org/10.1007/s11097-005-9002-y>
20. Lyon, P. The biogenic approach to cognition. *Cognitive Processing* **2006**, *7*, 11–29. <https://doi.org/10.1007/s10339-005-0016-8>
21. Keijzer, F.; van Duijn, M.; Lyon, P. What nervous systems do: early evolution, input–output, and the skin brain thesis. *Adaptive Behavior* **2013**, *21*, 67–85. <https://doi.org/10.1177/1059712312465330>
22. Levin, M. Bioelectric networks: the cognitive glue enabling evolutionary scaling from physiology to mind. *Animal Cognition* **2023**, *26*, 1865–1891. <https://doi.org/10.1007/s10071-023-01780-3>
23. Baluška, F.; Reber, A. Sentience and Consciousness in Single Cells: How the First Minds Emerged in Unicellular Species. *BioEssays* **2019**, *41*, 1800229. <https://doi.org/10.1002/bies.201800229>

24. Feinberg, T.E.; Mallatt, J. The evolutionary and genetic origins of consciousness in the Cambrian Period over 500 million years ago. *Frontiers in Psychology* **2013**, *4*. <https://doi.org/10.3389/fpsyg.2013.00667>
25. Ginsburg, S.; Jablonka, E. The evolution of associative learning: A factor in the Cambrian explosion. *Journal of Theoretical Biology* **2010**, *266*, 11–20. <https://doi.org/10.1016/j.jtbi.2010.06.017>
26. Damasio, A.; Carvalho, G.B. The nature of feelings: evolutionary and neurobiological origins. *Nature Reviews Neuroscience* **2013**, *14*, 143–152. <https://doi.org/10.1038/nrn3403>
27. Panksepp, J. Affective consciousness: Core emotional feelings in animals and humans. *Consciousness and Cognition* **2005**, *14*, 30–80. <https://doi.org/10.1016/j.concog.2004.10.004>
28. Solms, M. The Hard Problem of Consciousness and the Free Energy Principle. *Frontiers in Psychology* **2019**, *9*, 2714. <https://doi.org/10.3389/fpsyg.2018.02714>
29. Merker, B. Consciousness without a cerebral cortex: A challenge for neuroscience and medicine. *Behavioral and Brain Sciences* **2007**, *30*, 63–81. <https://doi.org/10.1017/s0140525x07000891>
30. New York Declaration on Animal Consciousness, 2024. Declaration issued at New York University, 19 April 2024. <https://sites.google.com/nyu.edu/nydeclaration/declaration> (accessed 24 September 2026).
31. Barron, A.B.; Halina, M.; Klein, C. Transitions in cognitive evolution. *Proceedings of the Royal Society B: Biological Sciences* **2023**, *290*, 20230671. <https://doi.org/10.1098/rspb.2023.0671>
32. Redgrave, P.; Prescott, T.J.; Gurney, K. The basal ganglia: a vertebrate solution to the selection problem?. *Neuroscience* **1999**, *89*, 1009–1023. [https://doi.org/10.1016/s0306-4522\(98\)00319-4](https://doi.org/10.1016/s0306-4522(98)00319-4)
33. Gurney, K.; Prescott, T.J.; Redgrave, P. A computational model of action selection in the basal ganglia. I. A new functional anatomy. *Biological Cybernetics* **2001**, *84*, 401–410. <https://doi.org/10.1007/pl00007984>
34. Redgrave, P.; Gurney, K. The short-latency dopamine signal: a role in discovering novel actions?. *Nature Reviews Neuroscience* **2006**, *7*, 967–975. <https://doi.org/10.1038/nrn2022>
35. Gold, J.I.; Shadlen, M.N. The Neural Basis of Decision Making. *Annual Review of Neuroscience* **2007**, *30*, 535–574. <https://doi.org/10.1146/annurev.neuro.29.051605.113038>
36. Cisek, P. Cortical mechanisms of action selection: the affordance competition hypothesis. *Philosophical Transactions of the Royal Society B: Biological Sciences* **2007**, *362*, 1585–1599. <https://doi.org/10.1098/rstb.2007.2054>
37. Cisek, P.; Kalaska, J.F. Neural Correlates of Reaching Decisions in Dorsal Premotor Cortex: Specification of Multiple Direction Choices and Final Selection of Action. *Neuron* **2005**, *45*, 801–814. <https://doi.org/10.1016/j.neuron.2005.01.027>
38. Morsella, E. The Function of Phenomenal States: Supramodular Interaction Theory. *Psychological Review* **2005**, *112*, 1000–1021. <https://doi.org/10.1037/0033-295x.112.4.1000>
39. Morsella, E.; Godwin, C.A.; Jantz, T.K.; Krieger, S.C.; Gazzaley, A. Homing in on consciousness in the nervous system: An action-based synthesis. *Behavioral and Brain Sciences* **2016**, *39*, e168. <https://doi.org/10.1017/s0140525x15000643>
40. Morsella, E.; Gray, J.R.; Krieger, S.C.; Bargh, J.A. The essence of conscious conflict: Subjective effects of sustaining incompatible intentions. *Emotion* **2009**, *9*, 717–728. <https://doi.org/10.1037/a0017121>
41. Gray, J.R.; Bargh, J.A.; Morsella, E. Neural correlates of the essence of conscious conflict: fMRI of sustaining incompatible intentions. *Experimental Brain Research* **2013**, *229*, 453–465. <https://doi.org/10.1007/s00221-013-3566-5>
42. Clark, K.B. Possible origins of consciousness in simple control over “involuntary” neuroimmunological action. *Consciousness and Cognition* **2018**, *61*, 76–78. <https://doi.org/10.1016/j.concog.2018.04.002>
43. Heredia Cedillo, A.; Lambert, D.; Morsella, E. Identifying Consciousness in Other Creatures: Three Initial Steps. *Behavioral Sciences* **2024**, *14*, 337. <https://doi.org/10.3390/bs14040337>
44. Cabanac, M. Pleasure: the common currency. *Journal of Theoretical Biology* **1992**, *155*, 173–200. [https://doi.org/10.1016/s0022-5193\(05\)80594-6](https://doi.org/10.1016/s0022-5193(05)80594-6)
45. Levy, D.J.; Glimcher, P.W. The root of all value: a neural common currency for choice. *Current Opinion in Neurobiology* **2012**, *22*, 1027–1038. <https://doi.org/10.1016/j.conb.2012.06.001>
46. Craig, A.D. How do you feel? Interoception: the sense of the physiological condition of the body. *Nature Reviews Neuroscience* **2002**, *3*, 655–666. <https://doi.org/10.1038/nrn894>
47. Seth, A.K. Interoceptive inference, emotion, and the embodied self. *Trends in Cognitive Sciences* **2013**, *17*, 565–573. <https://doi.org/10.1016/j.tics.2013.09.007>
48. Seth, A.K.; Friston, K.J. Active interoceptive inference and the emotional brain. *Philosophical Transactions of the Royal Society B: Biological Sciences* **2016**, *371*, 20160007. <https://doi.org/10.1098/rstb.2016.0007>

49. Sterling, P. Allostasis: A model of predictive regulation. *Physiology & Behavior* **2012**, *106*, 5–15. <https://doi.org/10.1016/j.physbeh.2011.06.004>
50. Kleckner, I.R.; Zhang, J.; Touroutoglou, A.; Chanes, L.; Xia, C.; Simmons, W.K.; Quigley, K.S.; Dickerson, B.C.; Feldman Barrett, L. Evidence for a large-scale brain system supporting allostasis and interoception in humans. *Nature Human Behaviour* **2017**, *1*, 0069. <https://doi.org/10.1038/s41562-017-0069>
51. Sennesh, E.; Theriault, J.; Brooks, D.; van de Meent, J.W.; Barrett, L.F.; Quigley, K.S. Interoception as modeling, allostasis as control. *Biological Psychology* **2022**, *167*, 108242. <https://doi.org/10.1016/j.biopsycho.2021.108242>
52. Scherer, K.R.; Moors, A. The Emotion Process: Event Appraisal and Component Differentiation. *Annual Review of Psychology* **2019**, *70*, 719–745. <https://doi.org/10.1146/annurev-psych-122216-011854>
53. Nesse, R.M. Evolutionary explanations of emotions. *Human Nature* **1990**, *1*, 261–289. <https://doi.org/10.1007/bf02733986>
54. Frijda, N.H. Emotions and Action. In *Feelings and Emotions*; Cambridge University Press, 2004; pp. 158–173. <https://doi.org/10.1017/cbo9780511806582.010>
55. Barrett, L.F. The theory of constructed emotion: an active inference account of interoception and categorization. *Social Cognitive and Affective Neuroscience* **2016**, nsw154. <https://doi.org/10.1093/scan/nsw154>
56. Lindquist, K.A.; Wager, T.D.; Kober, H.; Bliss-Moreau, E.; Barrett, L.F. The brain basis of emotion: A meta-analytic review. *Behavioral and Brain Sciences* **2012**, *35*, 121–143. <https://doi.org/10.1017/s0140525x11000446>
57. Siegel, E.H.; Sands, M.K.; Van den Noortgate, W.; Condon, P.; Chang, Y.; Dy, J.; Quigley, K.S.; Barrett, L.F. Emotion fingerprints or emotion populations? A meta-analytic investigation of autonomic features of emotion categories. *Psychological Bulletin* **2018**, *144*, 343–393. <https://doi.org/10.1037/bul0000128>
58. Barrett, L.F. Are Emotions Natural Kinds?. *Perspectives on Psychological Science* **2006**, *1*, 28–58. <https://doi.org/10.1111/j.1745-6916.2006.00003.x>
59. Joffily, M.; Coricelli, G. Emotional Valence and the Free-Energy Principle. *PLoS Computational Biology* **2013**, *9*, e1003094. <https://doi.org/10.1371/journal.pcbi.1003094>
60. Berridge, K.C.; Kringelbach, M.L. Pleasure Systems in the Brain. *Neuron* **2015**, *86*, 646–664. <https://doi.org/10.1016/j.neuron.2015.02.018>
61. LeDoux, J. Rethinking the Emotional Brain. *Neuron* **2012**, *73*, 653–676. <https://doi.org/10.1016/j.neuron.2012.02.004>
62. LeDoux, J.E.; Brown, R. A higher-order theory of emotional consciousness. *Proceedings of the National Academy of Sciences* **2017**, *114*. <https://doi.org/10.1073/pnas.1619316114>
63. Lau, H.; Rosenthal, D. Empirical support for higher-order theories of conscious awareness. *Trends in Cognitive Sciences* **2011**, *15*, 365–373. <https://doi.org/10.1016/j.tics.2011.05.009>
64. Sneddon, L.U. Comparative Physiology of Nociception and Pain. *Physiology* **2018**, *33*, 63–73. <https://doi.org/10.1152/physiol.00022.2017>
65. Sneddon, L.U.; Elwood, R.W.; Adamo, S.A.; Leach, M.C. Defining and assessing animal pain. *Animal Behaviour* **2014**, *97*, 201–212. <https://doi.org/10.1016/j.anbehav.2014.09.007>
66. Dehaene, S.; Changeux, J.P. Experimental and Theoretical Approaches to Conscious Processing. *Neuron* **2011**, *70*, 200–227. <https://doi.org/10.1016/j.neuron.2011.03.018>
67. Mashour, G.A.; Roelfsema, P.; Changeux, J.P.; Dehaene, S. Conscious Processing and the Global Neuronal Workspace Hypothesis. *Neuron* **2020**, *105*, 776–798. <https://doi.org/10.1016/j.neuron.2020.01.026>
68. Sergent, C.; Baillet, S.; Dehaene, S. Timing of the brain events underlying access to consciousness during the attentional blink. *Nature Neuroscience* **2005**, *8*, 1391–1400. <https://doi.org/10.1038/nn1549>
69. Del Cul, A.; Baillet, S.; Dehaene, S. Brain Dynamics Underlying the Nonlinear Threshold for Access to Consciousness. *PLoS Biology* **2007**, *5*, e260. <https://doi.org/10.1371/journal.pbio.0050260>
70. Shanahan, M.; Baars, B. Applying global workspace theory to the frame problem. *Cognition* **2005**, *98*, 157–176. <https://doi.org/10.1016/j.cognition.2004.11.007>
71. Dehaene, S.; Charles, L.; King, J.R.; Marti, S. Toward a computational theory of conscious processing. *Current Opinion in Neurobiology* **2014**, *25*, 76–84. <https://doi.org/10.1016/j.conb.2013.12.005>
72. Oizumi, M.; Albantakis, L.; Tononi, G. From the Phenomenology to the Mechanisms of Consciousness: Integrated Information Theory 3.0. *PLoS Computational Biology* **2014**, *10*, e1003588. <https://doi.org/10.1371/journal.pcbi.1003588>
73. Tononi, G.; Boly, M.; Massimini, M.; Koch, C. Integrated information theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience* **2016**, *17*, 450–461. <https://doi.org/10.1038/nrn.2016.44>

74. Albantakis, L.; Barbosa, L.; Findlay, G.; Grasso, M.; Haun, A.M.; Marshall, W.; Mayner, W.G.P.; Zaeemzadeh, A.; Boly, M.; Juel, B.E.; et al. Integrated information theory (IIT) 4.0: Formulating the properties of phenomenal existence in physical terms. *PLOS Computational Biology* **2023**, *19*, e1011465. <https://doi.org/10.1371/journal.pcbi.1011465>
75. Lamme, V.A.F.; Roelfsema, P.R. The distinct modes of vision offered by feedforward and recurrent processing. *Trends in Neurosciences* **2000**, *23*, 571–579. [https://doi.org/10.1016/s0166-2236\(00\)01657-x](https://doi.org/10.1016/s0166-2236(00)01657-x)
76. Lamme, V.A.F. Towards a true neural stance on consciousness. *Trends in Cognitive Sciences* **2006**, *10*, 494–501. <https://doi.org/10.1016/j.tics.2006.09.001>
77. Boly, M.; Massimini, M.; Tsuchiya, N.; Postle, B.R.; Koch, C.; Tononi, G. Are the Neural Correlates of Consciousness in the Front or in the Back of the Cerebral Cortex? Clinical and Neuroimaging Evidence. *The Journal of Neuroscience* **2017**, *37*, 9603–9613. <https://doi.org/10.1523/jneurosci.3218-16.2017>
78. Yaron, I.; Melloni, L.; Pitts, M.; Mudrik, L. The ConTraSt database for analysing and comparing empirical studies of consciousness theories. *Nature Human Behaviour* **2022**, *6*, 593–604. <https://doi.org/10.1038/s41562-021-01284-5>
79. Kuhn, R.L. A landscape of consciousness: Toward a taxonomy of explanations and implications. *Progress in Biophysics and Molecular Biology* **2024**, *190*, 28–169. <https://doi.org/10.1016/j.pbiomolbio.2023.12.003>
80. Poznanski, R.R. Processual relational geometry: generative unfolding and self-intending closure constituting consciousness — a transscale ontophysical framework. *Journal of Multiscale Neuroscience* **2026**, *5*, 30–53. <https://doi.org/10.56280/1744116056>
81. Brown, R.; Lau, H.; LeDoux, J.E. Understanding the Higher-Order Approach to Consciousness. *Trends in Cognitive Sciences* **2019**, *23*, 754–768. <https://doi.org/10.1016/j.tics.2019.06.009>
82. Fleming, S.M. Awareness as inference in a higher-order state space. *Neuroscience of Consciousness* **2020**, *2020*, niz020. <https://doi.org/10.1093/nc/niz020>
83. Dijkstra, N.; Fleming, S.M. Subjective signal strength distinguishes reality from imagination. *Nature Communications* **2023**, *14*, 1627. <https://doi.org/10.1038/s41467-023-37322-1>
84. Graziano, M.S.A.; Webb, T.W. The attention schema theory: a mechanistic account of subjective awareness. *Frontiers in Psychology* **2015**, *06*. <https://doi.org/10.3389/fpsyg.2015.00500>
85. Wilterson, A.I.; Kemper, C.M.; Kim, N.; Webb, T.W.; Reblando, A.M.W.; Graziano, M.S.A. Attention control and the attention schema theory of consciousness. *Progress in Neurobiology* **2020**, *195*, 101844. <https://doi.org/10.1016/j.pneurobio.2020.101844>
86. Graziano, M.S.A.; Guterstam, A.; Bio, B.J.; Wilterson, A.I. Toward a standard model of consciousness: Reconciling the attention schema, global workspace, higher-order thought, and illusionist theories. *Cognitive Neuropsychology* **2020**, *37*, 155–172. <https://doi.org/10.1080/02643294.2019.1670630>
87. Clark, A. Whatever next? Predictive brains, situated agents, and the future of cognitive science. *Behavioral and Brain Sciences* **2013**, *36*, 181–204. <https://doi.org/10.1017/s0140525x12000477>
88. Seth, A.K.; Tsakiris, M. Being a Beast Machine: The Somatic Basis of Selfhood. *Trends in Cognitive Sciences* **2018**, *22*, 969–981. <https://doi.org/10.1016/j.tics.2018.08.008>
89. Hohwy, J.; Seth, A. Predictive processing as a systematic basis for identifying the neural correlates of consciousness. *Philosophy and the Mind Sciences* **2020**, *1*. <https://doi.org/10.33735/phimisci.2020.ii.64>
90. Chalmers, D.J. Facing Up to the Problem of Consciousness. In *The Character of Consciousness*; Oxford University Press, 2010; pp. 3–34. <https://doi.org/10.1093/acprof:oso/9780195311105.003.0001>
91. Frankish, K. Illusionism. *Philosophers on Consciousness* **2022**. <https://doi.org/10.5040/9781350190450.ch-008>
92. Melloni, L.; Mudrik, L.; Pitts, M.; Koch, C. Making the hard problem of consciousness easier. *Science* **2021**, *372*, 911–912. <https://doi.org/10.1126/science.abj3259>
93. Doerig, A.; Schurger, A.; Hess, K.; Herzog, M.H. The unfolding argument: Why IIT and other causal structure theories cannot explain consciousness. *Consciousness and Cognition* **2019**, *72*, 49–59. <https://doi.org/10.1016/j.concog.2019.04.002>
94. Tsuchiya, N.; Andriillon, T.; Haun, A. A reply to “the unfolding argument”: Beyond functionalism/behaviorism and towards a science of causal structure theories of consciousness. *Consciousness and Cognition* **2020**, *79*, 102877. <https://doi.org/10.1016/j.concog.2020.102877>
95. Kleiner, J. Brain states matter. A reply to the unfolding argument. *Consciousness and Cognition* **2020**, *85*, 102981. <https://doi.org/10.1016/j.concog.2020.102981>

96. Kleiner, J.; Hoel, E. Falsification and consciousness. *Neuroscience of Consciousness* **2021**, *2021*, niab001. <https://doi.org/10.1093/nc/niab001>
97. Doerig, A.; Schurger, A.; Herzog, M.H. Hard criteria for empirical theories of consciousness. *Cognitive Neuroscience* **2021**, *12*, 41–62. <https://doi.org/10.1080/17588928.2020.1772214>
98. Negro, N. (Dis)confirming theories of consciousness and their predictions: towards a Lakatosian consciousness science. *Neuroscience of Consciousness* **2024**, *2024*, niae012. <https://doi.org/10.1093/nc/niae012>
99. Mediano, P.A.M.; Rosas, F.E.; Bor, D.; Seth, A.K.; Barrett, A.B. The strength of weak integrated information theory. *Trends in Cognitive Sciences* **2022**, *26*, 646–655. <https://doi.org/10.1016/j.tics.2022.04.008>
100. IIT-Concerned; Arnold, D.H.; Baxter, M.G.; Bekinschtein, T.A.; Bengio, Y.; Bisley, J.W.; Browning, J.; Buonomano, D.; Carmel, D.; Carrasco, M.; et al. What makes a theory of consciousness unscientific?. *Nature Neuroscience* **2025**, *28*, 689–693. <https://doi.org/10.1038/s41593-025-01881-x>
101. Tononi, G.; Albantakis, L.; Barbosa, L.; Boly, M.; Cirelli, C.; Comolatti, R.; Ellia, F.; Findlay, G.; Casali, A.G.; Grasso, M.; et al. Consciousness or pseudo-consciousness? A clash of two paradigms. *Nature Neuroscience* **2025**, *28*, 694–702. <https://doi.org/10.1038/s41593-025-01880-y>
102. Tsuchiya, N.; Wilke, M.; Frässle, S.; Lamme, V.A.F. No-Report Paradigms: Extracting the True Neural Correlates of Consciousness. *Trends in Cognitive Sciences* **2015**, *19*, 757–770. <https://doi.org/10.1016/j.tics.2015.10.002>
103. Pitts, M.A.; Metzler, S.; Hillyard, S.A. Isolating neural correlates of conscious perception from neural correlates of reporting one's perception. *Frontiers in Psychology* **2014**, *5*. <https://doi.org/10.3389/fpsyg.2014.01078>
104. Sergent, C.; Corazzol, M.; Labouret, G.; Stockart, F.; Wexler, M.; King, J.R.; Meyniel, F.; Pressnitzer, D. Bifurcation in brain dynamics reveals a signature of conscious processing independent of report. *Nature Communications* **2021**, *12*, 1149. <https://doi.org/10.1038/s41467-021-21393-z>
105. Kronemer, S.I.; Aksen, M.; Ding, J.Z.; Ryu, J.H.; Xin, Q.; Ding, Z.; Prince, J.S.; Kwon, H.; Khalaf, A.; Forman, S.; et al. Human visual consciousness involves large scale cortical and subcortical networks independent of task report and eye movement activity. *Nature Communications* **2022**, *13*, 7342. <https://doi.org/10.1038/s41467-022-35117-4>
106. Dellert, T.; Balster, H.; Schlossmacher, I.; Bruchmann, M.; Moeck, R.; Straube, T. Neural correlates of consciousness in an auditory no-report fMRI study. *Current Biology* **2025**, *35*, 5721–5733.e3. <https://doi.org/10.1016/j.cub.2025.10.026>
107. Odegaard, B.; Knight, R.T.; Lau, H. Should a few null findings falsify prefrontal theories of conscious perception?. *bioRxiv* **2017**, 122267 (preprint). <https://doi.org/10.1101/122267>
108. Block, N. What Is Wrong with the No-Report Paradigm and How to Fix It. *Trends in Cognitive Sciences* **2019**, *23*, 1003–1013. <https://doi.org/10.1016/j.tics.2019.10.001>
109. Aru, J.; Suzuki, M.; Larkum, M.E. Cellular Mechanisms of Conscious Processing. *Trends in Cognitive Sciences* **2020**, *24*, 814–825. <https://doi.org/10.1016/j.tics.2020.07.006>
110. Suzuki, M.; Larkum, M.E. General Anesthesia Decouples Cortical Pyramidal Neurons. *Cell* **2020**, *180*, 666–676.e13. <https://doi.org/10.1016/j.cell.2020.01.024>
111. Northoff, G.; Huang, Z. How do the brain's time and space mediate consciousness and its different dimensions? Temporo-spatial theory of consciousness (TTC). *Neuroscience & Biobehavioral Reviews* **2017**, *80*, 630–645. <https://doi.org/10.1016/j.neubiorev.2017.07.013>
112. Northoff, G.; Zilio, F. Temporo-spatial Theory of Consciousness (TTC) – Bridging the gap of neuronal activity and phenomenal states. *Behavioural Brain Research* **2022**, *424*, 113788. <https://doi.org/10.1016/j.bbr.2022.113788>
113. Whyte, C.J.; Smith, R. The predictive global neuronal workspace: A formal active inference model of visual consciousness. *Progress in Neurobiology* **2021**, *199*, 101918. <https://doi.org/10.1016/j.pneurobio.2020.101918>
114. Ginsburg, S.; Jablonka, E. *The Evolution of the Sensitive Soul*; The MIT Press, 2019. <https://doi.org/10.7551/mitpress/11006.001.0001>
115. Birch, J.; Ginsburg, S.; Jablonka, E. Unlimited Associative Learning and the origins of consciousness: a primer and some predictions. *Biology & Philosophy* **2020**, *35*, 56. <https://doi.org/10.1007/s10539-020-09772-0>
116. Rounis, E.; Maniscalco, B.; Rothwell, J.C.; Passingham, R.E.; Lau, H. Theta-burst transcranial magnetic stimulation to the prefrontal cortex impairs metacognitive visual awareness. *Cognitive Neuroscience* **2010**, *1*, 165–175. <https://doi.org/10.1080/17588921003632529>
117. Graziano, M.S.A.; Kastner, S. Human consciousness and its relationship to social neuroscience: A novel hypothesis. *Cognitive Neuroscience* **2011**, *2*, 98–113. <https://doi.org/10.1080/17588928.2011.565121>

118. Garfinkel, S.N.; Seth, A.K.; Barrett, A.B.; Suzuki, K.; Critchley, H.D. Knowing your own heart: Distinguishing interoceptive accuracy from interoceptive awareness. *Biological Psychology* **2015**, *104*, 65–74. <https://doi.org/10.1016/j.biopsycho.2014.11.004>
119. Park, H.D.; Tallon-Baudry, C. The neural subjective frame: from bodily signals to perceptual consciousness. *Philosophical Transactions of the Royal Society B: Biological Sciences* **2014**, *369*, 20130208. <https://doi.org/10.1098/rstb.2013.0208>
120. Park, H.D.; Correia, S.; Ducorps, A.; Tallon-Baudry, C. Spontaneous fluctuations in neural responses to heartbeats predict visual detection. *Nature Neuroscience* **2014**, *17*, 612–618. <https://doi.org/10.1038/nn.3671>
121. Dehaene, S.; Changeux, J.P.; Naccache, L.; Sackur, J.; Sergent, C. Conscious, preconscious, and subliminal processing: a testable taxonomy. *Trends in Cognitive Sciences* **2006**, *10*, 204–211. <https://doi.org/10.1016/j.tics.2006.03.007>
122. Al, E.; Iliopoulos, F.; Forschack, N.; Nierhaus, T.; Grund, M.; Motyka, P.; Gaebler, M.; Nikulin, V.V.; Villringer, A. Heart–brain interactions shape somatosensory perception and evoked potentials. *Proceedings of the National Academy of Sciences* **2020**, *117*, 10575–10584. <https://doi.org/10.1073/pnas.1915629117>
123. Al, E.; Iliopoulos, F.; Nikulin, V.V.; Villringer, A. Heartbeat and somatosensory perception. *NeuroImage* **2021**, *238*, 118247. <https://doi.org/10.1016/j.neuroimage.2021.118247>
124. Kluger, D.S.; Balestrieri, E.; Busch, N.A.; Gross, J. Respiration aligns perception with neural excitability. *eLife* **2021**, *10*, e70907. <https://doi.org/10.7554/elife.70907>
125. Grund, M.; Al, E.; Pabst, M.; Dabbagh, A.; Stephani, T.; Nierhaus, T.; Gaebler, M.; Villringer, A. Respiration, Heartbeat, and Conscious Tactile Perception. *The Journal of Neuroscience* **2022**, *42*, 643–656. <https://doi.org/10.1523/jneurosci.0592-21.2021>
126. Candia-Rivera, D.; Annen, J.; Gosseries, O.; Martial, C.; Thibaut, A.; Laureys, S.; Tallon-Baudry, C. Neural Responses to Heartbeats Detect Residual Signs of Consciousness during Resting State in Postcomatose Patients. *The Journal of Neuroscience* **2021**, *41*, 5251–5262. <https://doi.org/10.1523/jneurosci.1740-20.2021>
127. Tallon-Baudry, C.; Campana, F.; Park, H.D.; Babo-Rebello, M. The neural monitoring of visceral inputs, rather than attention, accounts for first-person perspective in conscious vision. *Cortex* **2018**, *102*, 139–149. <https://doi.org/10.1016/j.cortex.2017.05.019>
128. Morsella, E.; Wilson, L.E.; Berger, C.C.; Honnhongva, M.; Gazzaley, A.; Bargh, J.A. Subjective aspects of cognitive control at different stages of processing. *Attention, Perception & Psychophysics* **2009**, *71*, 1807–1824. <https://doi.org/10.3758/app.71.8.1807>
129. Desender, K.; Van Opstal, F.; Van den Bussche, E. Feeling the Conflict. *Psychological Science* **2014**, *25*, 675–683. <https://doi.org/10.1177/0956797613511468>
130. Yang, E.; Zald, D.H.; Blake, R. Fearful expressions gain preferential access to awareness during continuous flash suppression. *Emotion* **2007**, *7*, 882–886. <https://doi.org/10.1037/1528-3542.7.4.882>
131. Whalen, P.J.; Rauch, S.L.; Etcoff, N.L.; McInerney, S.C.; Lee, M.B.; Jenike, M.A. Masked Presentations of Emotional Facial Expressions Modulate Amygdala Activity without Explicit Knowledge. *The Journal of Neuroscience* **1998**, *18*, 411–418. <https://doi.org/10.1523/jneurosci.18-01-00411.1998>
132. Farb, N.A.S.; Segal, Z.V.; Anderson, A.K. Mindfulness meditation training alters cortical representations of interoceptive attention. *Social Cognitive and Affective Neuroscience* **2013**, *8*, 15–26. <https://doi.org/10.1093/scan/nss066>
133. Sheth, B.R.; Pham, T. How emotional arousal and valence influence access to awareness. *Vision Research* **2008**, *48*, 2415–2424. <https://doi.org/10.1016/j.visres.2008.07.013>
134. Gayet, S.; Paffen, C.L.E.; Belopolsky, A.V.; Theeuwes, J.; Van der Stigchel, S. Visual input signaling threat gains preferential access to awareness in a breaking continuous flash suppression paradigm. *Cognition* **2016**, *149*, 77–83. <https://doi.org/10.1016/j.cognition.2016.01.009>
135. Maniscalco, B.; Lau, H. A signal detection theoretic approach for estimating metacognitive sensitivity from confidence ratings. *Consciousness and Cognition* **2012**, *21*, 422–430. <https://doi.org/10.1016/j.concog.2011.09.021>
136. Fleming, S.M.; Lau, H.C. How to measure metacognition. *Frontiers in Human Neuroscience* **2014**, *8*. <https://doi.org/10.3389/fnhum.2014.00443>
137. Bor, D.; Schwartzman, D.J.; Barrett, A.B.; Seth, A.K. Theta-burst transcranial magnetic stimulation to the prefrontal or parietal cortex does not impair metacognitive visual awareness. *PLOS ONE* **2017**, *12*, e0171793. <https://doi.org/10.1371/journal.pone.0171793>

138. Ruby, E.; Maniscalco, B.; Peters, M.A.K. On a ‘failed’ attempt to manipulate visual metacognition with transcranial magnetic stimulation to prefrontal cortex. *Consciousness and Cognition* **2018**, *62*, 34–41. <https://doi.org/10.1016/j.concog.2018.04.009>
139. Bor, D.; Barrett, A.B.; Schwartzman, D.J.; Seth, A.K. Response to Ruby et al: On a ‘failed’ attempt to manipulate conscious perception with transcranial magnetic stimulation to prefrontal cortex. *Consciousness and Cognition* **2018**, *65*, 334–341. <https://doi.org/10.1016/j.concog.2018.07.011>
140. Fleming, S.M.; Weil, R.S.; Nagy, Z.; Dolan, R.J.; Rees, G. Relating Introspective Accuracy to Individual Differences in Brain Structure. *Science* **2010**, *329*, 1541–1543. <https://doi.org/10.1126/science.1191883>
141. Fleming, S.M.; Ryu, J.; Golfinos, J.G.; Blackmon, K.E. Domain-specific impairment in metacognitive accuracy following anterior prefrontal lesions. *Brain* **2014**, *137*, 2811–2822. <https://doi.org/10.1093/brain/awu221>
142. Vaccaro, A.G.; Fleming, S.M. Thinking about thinking: A coordinate-based meta-analysis of neuroimaging studies of metacognitive judgements. *Brain and Neuroscience Advances* **2018**, *2*, 2398212818810591. <https://doi.org/10.1177/2398212818810591>
143. Fleming, S.M. Metacognition and Confidence: A Review and Synthesis. *Annual Review of Psychology* **2024**, *75*, 241–268. <https://doi.org/10.1146/annurev-psych-022423-032425>
144. Desender, K.; Boldt, A.; Yeung, N. Subjective Confidence Predicts Information Seeking in Decision Making. *Psychological Science* **2018**, *29*, 761–778. <https://doi.org/10.1177/0956797617744771>
145. Rollwage, M.; Dolan, R.J.; Fleming, S.M. Metacognitive Failure as a Feature of Those Holding Radical Beliefs. *Current Biology* **2018**, *28*, 4014–4021.e8. <https://doi.org/10.1016/j.cub.2018.10.053>
146. Rouault, M.; Seow, T.; Gillan, C.M.; Fleming, S.M. Psychiatric Symptom Dimensions Are Associated With Dissociable Shifts in Metacognition but Not Task Performance. *Biological Psychiatry* **2018**, *84*, 443–451. <https://doi.org/10.1016/j.biopsych.2017.12.017>
147. Shea, N.; Boldt, A.; Bang, D.; Yeung, N.; Heyes, C.; Frith, C.D. Supra-personal cognitive control and metacognition. *Trends in Cognitive Sciences* **2014**, *18*, 186–193. <https://doi.org/10.1016/j.tics.2014.01.006>
148. Heyes, C.; Bang, D.; Shea, N.; Frith, C.D.; Fleming, S.M. Knowing Ourselves Together: The Cultural Origins of Metacognition. *Trends in Cognitive Sciences* **2020**, *24*, 349–362. <https://doi.org/10.1016/j.tics.2020.02.007>
149. Carruthers, P. How we know our own minds: The relationship between mindreading and metacognition. *Behavioral and Brain Sciences* **2009**, *32*, 121–138. <https://doi.org/10.1017/s0140525x09000545>
150. Shea, N.; Frith, C.D. The Global Workspace Needs Metacognition. *Trends in Cognitive Sciences* **2019**, *23*, 560–571. <https://doi.org/10.1016/j.tics.2019.04.007>
151. Cleeremans, A. The Radical Plasticity Thesis: How the Brain Learns to be Conscious. *Frontiers in Psychology* **2011**, *2*. <https://doi.org/10.3389/fpsyg.2011.00086>
152. Cleeremans, A.; Achoui, D.; Beauny, A.; Keuninckx, L.; Martin, J.R.; Muñoz-Moldes, S.; Vuillaume, L.; de Heering, A. Learning to Be Conscious. *Trends in Cognitive Sciences* **2020**, *24*, 112–123. <https://doi.org/10.1016/j.tics.2019.11.011>
153. Cleeremans, A.; Tallon-Baudry, C. Consciousness matters: phenomenal experience has functional value. *Neuroscience of Consciousness* **2022**, *2022*, niac007. <https://doi.org/10.1093/nc/niac007>
154. Ramsøy, T.Z.; Overgaard, M. Introspection and subliminal perception. *Phenomenology and the Cognitive Sciences* **2004**, *3*, 1–23. <https://doi.org/10.1023/b:phen.0000041900.30172.e8>
155. Sandberg, K.; Timmermans, B.; Overgaard, M.; Cleeremans, A. Measuring consciousness: Is one measure better than the other?. *Consciousness and Cognition* **2010**, *19*, 1069–1078. <https://doi.org/10.1016/j.concog.2009.12.013>
156. Overgaard, M.; Rote, J.; Mouridsen, K.; Ramsøy, T.Z. Is conscious perception gradual or dichotomous? A comparison of report methodologies during a visual task. *Consciousness and Cognition* **2006**, *15*, 700–708. <https://doi.org/10.1016/j.concog.2006.04.002>
157. Dehaene, S. Towards a cognitive neuroscience of consciousness: basic evidence and a workspace framework. *Cognition* **2001**, *79*, 1–37. [https://doi.org/10.1016/s0010-0277\(00\)00123-2](https://doi.org/10.1016/s0010-0277(00)00123-2)
158. Michel, M. The Mismeasure of Consciousness: A Problem of Coordination for the Perceptual Awareness Scale. *Philosophy of Science* **2019**, *86*, 1239–1249. <https://doi.org/10.1086/705509>
159. Lee, S.W.; Shimojo, S.; O’Doherty, J.P. Neural Computations Underlying Arbitration between Model-Based and Model-free Learning. *Neuron* **2014**, *81*, 687–699. <https://doi.org/10.1016/j.neuron.2013.11.028>
160. Boureau, Y.L.; Sokol-Hessner, P.; Daw, N.D. Deciding How To Decide: Self-Control and Meta-Decision Making. *Trends in Cognitive Sciences* **2015**, *19*, 700–710. <https://doi.org/10.1016/j.tics.2015.08.013>

161. Kool, W.; Botvinick, M. Mental labour. *Nature Human Behaviour* **2018**, *2*, 899–908. <https://doi.org/10.1038/s41562-018-0401-9>
162. Eppinger, B.; Goschke, T.; Musslick, S. Meta-control: From psychology to computational neuroscience. *Cognitive, Affective, & Behavioral Neuroscience* **2021**, *21*, 447–452. <https://doi.org/10.3758/s13415-021-00919-4>
163. Shenhav, A.; Botvinick, M.M.; Cohen, J.D. The Expected Value of Control: An Integrative Theory of Anterior Cingulate Cortex Function. *Neuron* **2013**, *79*, 217–240. <https://doi.org/10.1016/j.neuron.2013.07.007>
164. Shenhav, A.; Musslick, S.; Lieder, F.; Kool, W.; Griffiths, T.L.; Cohen, J.D.; Botvinick, M.M. Toward a Rational and Mechanistic Account of Mental Effort. *Annual Review of Neuroscience* **2017**, *40*, 99–124. <https://doi.org/10.1146/annurev-neuro-072116-031526>
165. Koechlin, E.; Ody, C.; Kouneiher, F. The Architecture of Cognitive Control in the Human Prefrontal Cortex. *Science* **2003**, *302*, 1181–1185. <https://doi.org/10.1126/science.1088545>
166. Musslick, S.; Cohen, J.D. Rationalizing constraints on the capacity for cognitive control. *Trends in Cognitive Sciences* **2021**, *25*, 757–775. <https://doi.org/10.1016/j.tics.2021.06.001>
167. Hagger, M.S.; Chatzisarantis, N.L.D.; Alberts, H.; Anggono, C.O.; Batailler, C.; Birt, A.R.; Brand, R.; Brandt, M.J.; Brewer, G.; Bruyneel, S.; et al. A Multilab Preregistered Replication of the Ego-Depletion Effect. *Perspectives on Psychological Science* **2016**, *11*, 546–573. <https://doi.org/10.1177/1745691616652873>
168. Vohs, K.D.; Schmeichel, B.J.; Lohmann, S.; Gronau, Q.F.; Finley, A.J.; Ainsworth, S.E.; Alquist, J.L.; Baker, M.D.; Brizi, A.; Bunyi, A.; et al. A Multisite Preregistered Paradigmatic Test of the Ego-Depletion Effect. *Psychological Science* **2021**, *32*, 1566–1581. <https://doi.org/10.1177/0956797621989733>
169. Kurzban, R.; Duckworth, A.; Kable, J.W.; Myers, J. An opportunity cost model of subjective effort and task performance. *Behavioral and Brain Sciences* **2013**, *36*, 661–679. <https://doi.org/10.1017/s0140525x12003196>
170. Logan, G.D.; Cowan, W.B. On the ability to inhibit thought and action: A theory of an act of control. *Psychological Review* **1984**, *91*, 295–327. <https://doi.org/10.1037/0033-295x.91.3.295>
171. Verbruggen, F.; Logan, G.D. Response inhibition in the stop-signal paradigm. *Trends in Cognitive Sciences* **2008**, *12*, 418–424. <https://doi.org/10.1016/j.tics.2008.07.005>
172. van Gaal, S.; Ridderinkhof, K.R.; Fahrenfort, J.J.; Scholte, H.S.; Lamme, V.A.F. Frontal Cortex Mediates Unconsciously Triggered Inhibitory Control. *The Journal of Neuroscience* **2008**, *28*, 8053–8062. <https://doi.org/10.1523/jneurosci.1278-08.2008>
173. van Gaal, S.; Ridderinkhof, K.R.; Scholte, H.S.; Lamme, V.A.F. Unconscious Activation of the Prefrontal No-Go Network. *The Journal of Neuroscience* **2010**, *30*, 4143–4150. <https://doi.org/10.1523/jneurosci.2992-09.2010>
174. van Gaal, S.; Lamme, V.A.F.; Fahrenfort, J.J.; Ridderinkhof, K.R. Dissociable Brain Mechanisms Underlying the Conscious and Unconscious Control of Behavior. *Journal of Cognitive Neuroscience* **2011**, *23*, 91–105. <https://doi.org/10.1162/jocn.2010.21431>
175. Botvinick, M.M.; Braver, T.S.; Barch, D.M.; Carter, C.S.; Cohen, J.D. Conflict monitoring and cognitive control. *Psychological Review* **2001**, *108*, 624–652. <https://doi.org/10.1037/0033-295x.108.3.624>
176. Brass, M.; Haggard, P. To Do or Not to Do: The Neural Signature of Self-Control. *The Journal of Neuroscience* **2007**, *27*, 9141–9145. <https://doi.org/10.1523/jneurosci.0924-07.2007>
177. Filevich, E.; Kühn, S.; Haggard, P. Intentional inhibition in human action: The power of ‘no’. *Neuroscience & Biobehavioral Reviews* **2012**, *36*, 1107–1118. <https://doi.org/10.1016/j.neubiorev.2012.01.006>
178. Klein, G.; Wolf, S.; Militello, L.; Zsombok, C. Characteristics of Skilled Option Generation in Chess. *Organizational Behavior and Human Decision Processes* **1995**, *62*, 63–69. <https://doi.org/10.1006/obhd.1995.1031>
179. Johnson, J.G.; Raab, M. Take The First: Option-generation and resulting choices. *Organizational Behavior and Human Decision Processes* **2003**, *91*, 215–229. [https://doi.org/10.1016/s0749-5978\(03\)00027-x](https://doi.org/10.1016/s0749-5978(03)00027-x)
180. Kaiser, S.; Simon, J.J.; Kalis, A.; Schweizer, S.; Tobler, P.N.; Mojzisch, A. The cognitive and neural basis of option generation and subsequent choice. *Cognitive, Affective, & Behavioral Neuroscience* **2013**, *13*, 814–829. <https://doi.org/10.3758/s13415-013-0175-5>
181. Huber, J.; Payne, J.W.; Puto, C. Adding Asymmetrically Dominated Alternatives: Violations of Regularity and the Similarity Hypothesis. *Journal of Consumer Research* **1982**, *9*, 90. <https://doi.org/10.1086/208899> [CrossRef lists the first page only; end page to be confirmed at production]
182. Thura, D.; Cisek, P. Deliberation and Commitment in the Premotor and Primary Motor Cortex during Dynamic Decision Making. *Neuron* **2014**, *81*, 1401–1416. <https://doi.org/10.1016/j.neuron.2014.01.031>
183. Nisbett, R.E.; Wilson, T.D. Telling more than we can know: Verbal reports on mental processes. *Psychological Review* **1977**, *84*, 231–259. <https://doi.org/10.1037/0033-295x.84.3.231>

184. Schwitzgebel, E. The Unreliability of Naive Introspection. *The Philosophical Review* **2008**, *117*, 245–273. <https://doi.org/10.1215/00318108-2007-037>
185. Johansson, P.; Hall, L.; Sikström, S.; Olsson, A. Failure to Detect Mismatches Between Intention and Outcome in a Simple Decision Task. *Science* **2005**, *310*, 116–119. <https://doi.org/10.1126/science.1111709>
186. Hall, L.; Johansson, P.; Tärning, B.; Sikström, S.; Deutgen, T. Magic at the marketplace: Choice blindness for the taste of jam and the smell of tea. *Cognition* **2010**, *117*, 54–61. <https://doi.org/10.1016/j.cognition.2010.06.010>
187. Hall, L.; Johansson, P.; Strandberg, T. Lifting the Veil of Morality: Choice Blindness and Attitude Reversals on a Self-Transforming Survey. *PLoS ONE* **2012**, *7*, e45457. <https://doi.org/10.1371/journal.pone.0045457>
188. Wegner, D.M.; Wheatley, T. Apparent mental causation: Sources of the experience of will. *American Psychologist* **1999**, *54*, 480–492. <https://doi.org/10.1037/0003-066x.54.7.480>
189. Moore, J.; Haggard, P. Awareness of action: Inference and prediction. *Consciousness and Cognition* **2008**, *17*, 136–144. <https://doi.org/10.1016/j.concog.2006.12.004>
190. Kühn, S.; Brass, M. Retrospective construction of the judgement of free choice. *Consciousness and Cognition* **2009**, *18*, 12–21. <https://doi.org/10.1016/j.concog.2008.09.007>
191. Haggard, P. Sense of agency in the human brain. *Nature Reviews Neuroscience* **2017**, *18*, 196–207. <https://doi.org/10.1038/nrn.2017.14>
192. Suzuki, K.; Lush, P.; Seth, A.K.; Roseboom, W. Intentional Binding Without Intentional Action. *Psychological Science* **2019**, *30*, 842–853. <https://doi.org/10.1177/0956797619842191>
193. Lush, P. Demand Characteristics Confound the Rubber Hand Illusion. *Collabra: Psychology* **2020**, *6*, 22. <https://doi.org/10.1525/collabra.325>
194. Libet, B.; Gleason, C.A.; Wright, E.W.; Pearl, D.K. Time of conscious intention to act in relation to onset of cerebral activity (readiness-potential). *Brain* **1983**, *106*, 623–642. <https://doi.org/10.1093/brain/106.3.623>
195. Schurger, A.; Sitt, J.D.; Dehaene, S. An accumulator model for spontaneous neural activity prior to self-initiated movement. *Proceedings of the National Academy of Sciences* **2012**, *109*. <https://doi.org/10.1073/pnas.1210467109>
196. Schurger, A.; Hu, P.; Pak, J.; Roskies, A.L. What Is the Readiness Potential?. *Trends in Cognitive Sciences* **2021**, *25*, 558–570. <https://doi.org/10.1016/j.tics.2021.04.001>
197. Metzinger, T. *Being No One*; The MIT Press, 2003. <https://doi.org/10.7551/mitpress/1551.001.0001>
198. Metzinger, T. The myth of cognitive agency: subpersonal thinking as a cyclically recurring loss of mental autonomy. *Frontiers in Psychology* **2013**, *4*. <https://doi.org/10.3389/fpsyg.2013.00931>
199. Limanowski, J.; Friston, K. ‘Seeing the Dark’: Grounding Phenomenal Transparency and Opacity in Precision Estimation for Active Inference. *Frontiers in Psychology* **2018**, *9*, 643. <https://doi.org/10.3389/fpsyg.2018.00643>
200. Cohen, M.A.; Ortego, K.; Kyroudis, A.; Pitts, M. Distinguishing the Neural Correlates of Perceptual Awareness and Postperceptual Processing. *The Journal of Neuroscience* **2020**, *40*, 4925–4935. <https://doi.org/10.1523/jneurosci.0120-20.2020>
201. Schlossmacher, I.; Dellert, T.; Pitts, M.; Bruchmann, M.; Straube, T. Differential Effects of Awareness and Task Relevance on Early and Late ERPs in a No-Report Visual Oddball Paradigm. *The Journal of Neuroscience* **2020**, *40*, 2906–2913. <https://doi.org/10.1523/jneurosci.2077-19.2020>
202. Dellert, T.; Müller-Bardorff, M.; Schlossmacher, I.; Pitts, M.; Hofmann, D.; Bruchmann, M.; Straube, T. Dissociating the Neural Correlates of Consciousness and Task Relevance in Face Perception Using Simultaneous EEG-fMRI. *The Journal of Neuroscience* **2021**, *41*, 7864–7875. <https://doi.org/10.1523/jneurosci.2799-20.2021>
203. Lakens, D.; Scheel, A.M.; Isager, P.M. Equivalence Testing for Psychological Research: A Tutorial. *Advances in Methods and Practices in Psychological Science* **2018**, *1*, 259–269. <https://doi.org/10.1177/2515245918770963>
204. Petzschner, F.H.; Weber, L.A.; Wellstein, K.V.; Paolini, G.; Do, C.T.; Stephan, K.E. Focus of attention modulates the heartbeat evoked potential. *NeuroImage* **2019**, *186*, 595–606. <https://doi.org/10.1016/j.neuroimage.2018.11.037>
205. Coll, M.P.; Hobson, H.; Bird, G.; Murphy, J. Systematic review and meta-analysis of the relationship between the heartbeat-evoked potential and interoception. *Neuroscience & Biobehavioral Reviews* **2021**, *122*, 190–200. <https://doi.org/10.1016/j.neubiorev.2020.12.012>